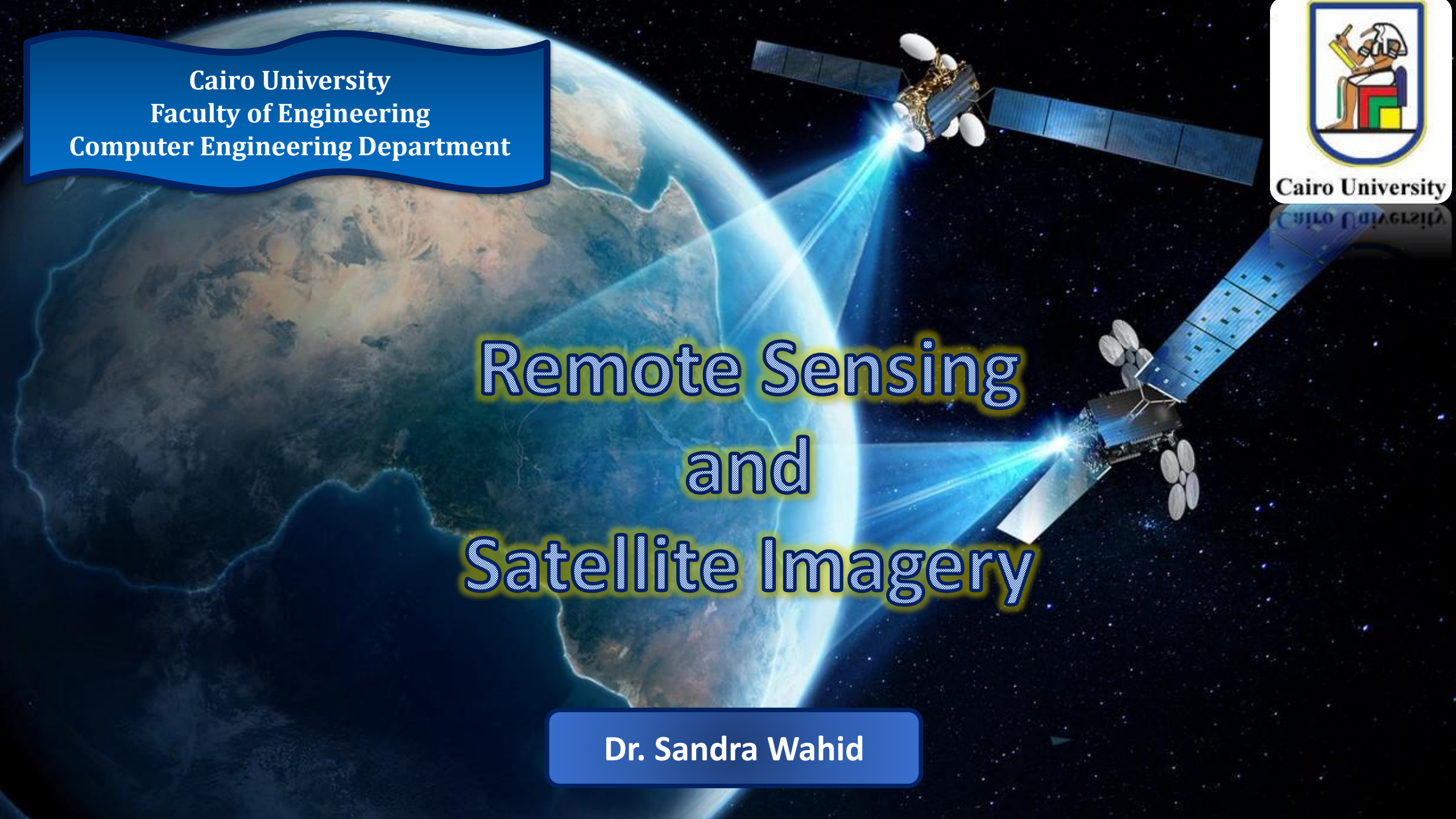
Cairo University
Faculty of Engineering
Computer Engineering Department



Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Remote Sensing

- Remote sensing is the process of detecting and monitoring the physical characteristics of an area by measuring its reflected and emitted radiation at a distance (typically from satellite or aircraft).

- Remote sensing is the acquiring of information from a distance. NASA observes Earth and other planetary bodies via remote sensors on satellites and aircraft that detect, and record reflected or emitted energy. Remote sensors, which provide a global perspective and a wealth of data about Earth systems, enable data-informed decision making based on the current and future state of our planet.

- In remote sensing, energy spreading out from the earth's surface is measured using a sensor mounted on an aircraft or spacecraft platform. That measurement is used to construct an image of the landscape beneath the platform.

Remote Sensing

- Provided an **energy source** is available, almost any wavelength could be used to image the characteristics of the earth's surface.

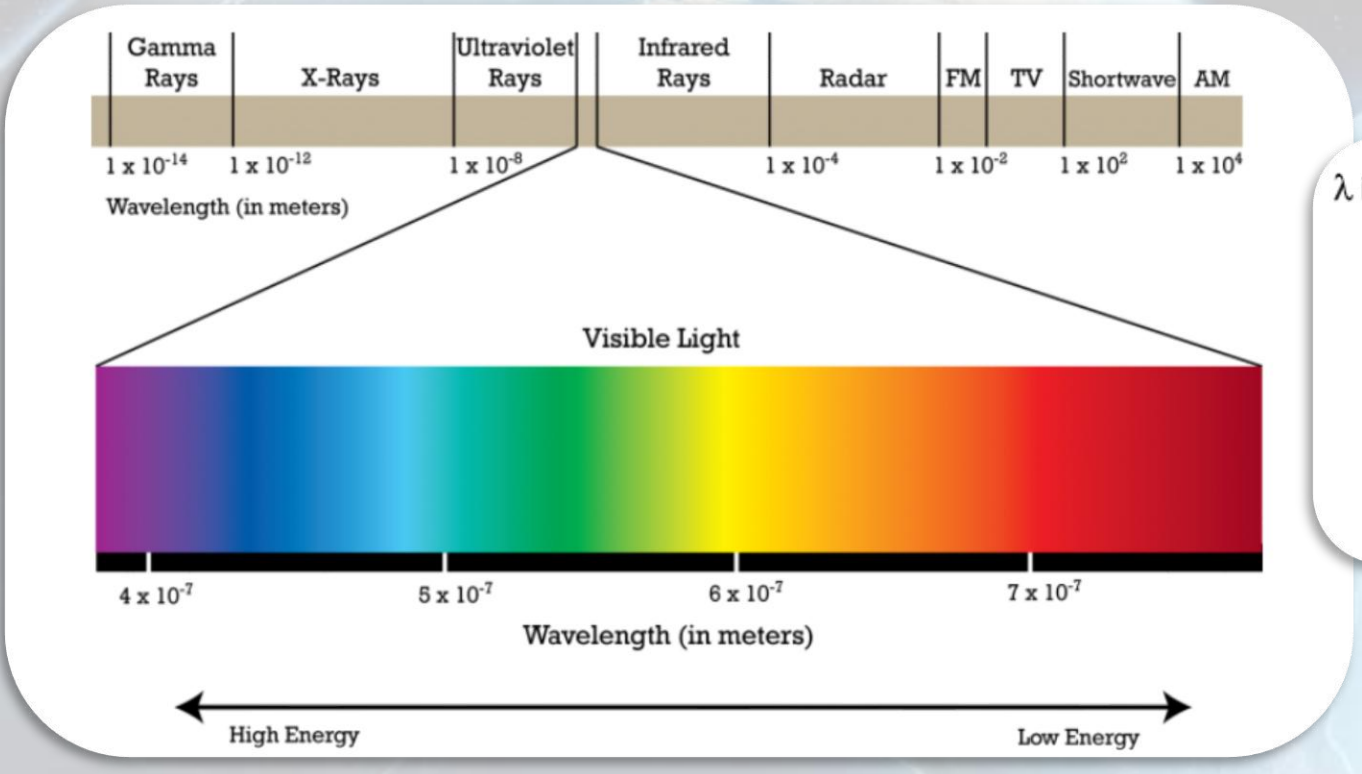
- **Sensors types:**

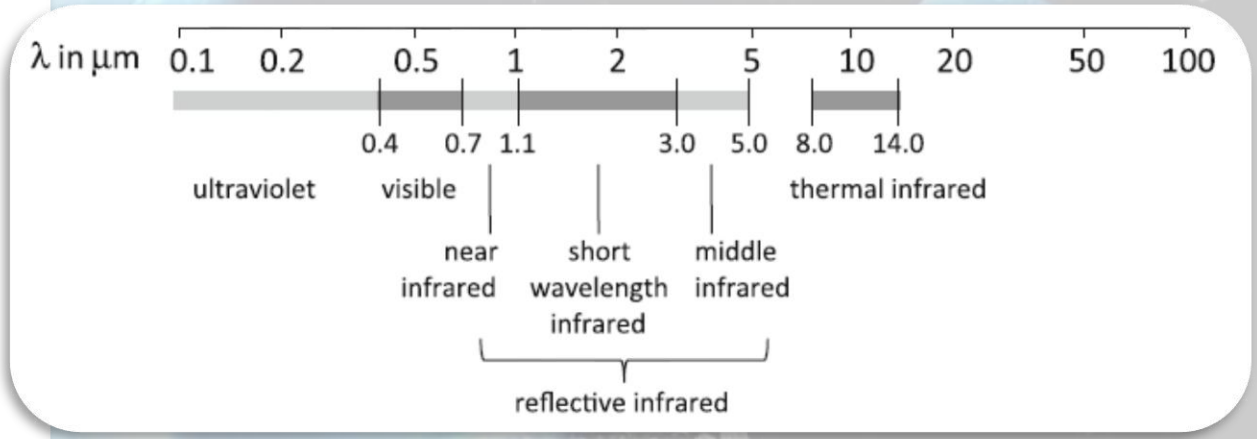
  - **Passive:** The sensors gather radiation that is emitted or reflected by the object or surrounding areas. Sunlight reflection is the most common source of radiation measured by passive sensors.

  - **Active:** The sensor embodies within itself the source of energy radiation. Energy is radiated from a platform onto the earth's surface. It is by measuring the energy scattered back to the platform that image data is recorded. For example, since earth emits too small microwave radiation to be measured, active sensors are used to record images at microwave wavelengths.

Energy Sources and Wavelength Ranges

- The electromagnetic spectrum:





IR divisions have variants

- The electromagnetic spectrum is broad and not all wavelengths are equally effective nor significantly interact with surfaces of interest for remote sensing purposes.
- A satellite image is created by combining measurements of the intensity of certain wavelengths of light, both visible and invisible to humans.
  - When we combine measurements of visible light, the resulting image is true color, or similar to what our eyes would see.
  - When we use non-visible light (usually infrared measurements), the resulting image is false color, and things might look different than we'd expect.

Applications of Remote Sensing

- **Land Cover and Land Use Mapping:**
  - Land Cover Mapping: represents information on different types (classes) of physical coverage of the Earth's surface, e.g. forests, grasslands, croplands, lakes, wetlands, …etc. Dynamic land cover maps include transitions of land cover classes over time and hence captures land cover changes.
  - Land Use Mapping: refers to the purpose the land serves, for example, housing, wildlife habitat, agriculture, …etc.

- **Weather Forecasting**

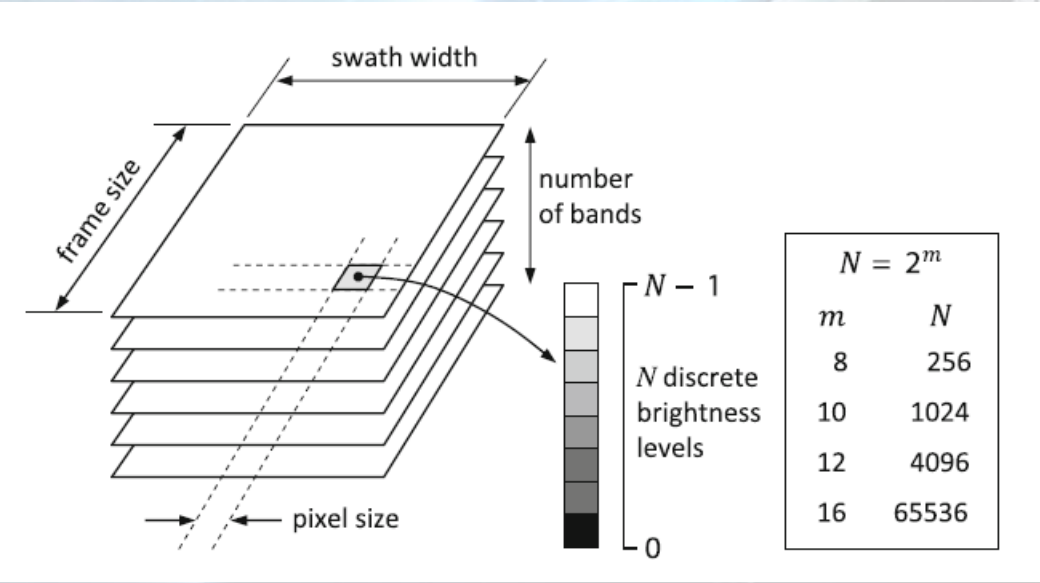
- **Environmental Study:** deforestation, degradation of fertile lands, pollution in atmosphere, desertification, oil spillage from oil tankers, …etc.

- **Study of Natural Hazards:** used to study damages caused by earthquakes, volcanoes, landslides, floods and melting of ice in polar regions. Also, remote sensing is helpful to predict the occurrence of natural hazards.

- **Resource Exploration:** updating existing geological maps, identifying the sites for the minerals and helpful in locating fuel deposits.

Primary Data Characteristics

- **Spectral resolution:** is the number and width of the spectral ***bands or channels*** that the satellite sensor detects.
- **Spatial resolution:** is described by the ***pixel size***.
- **Radiometric resolution:** is expressed in terms of the number of binary digits, or bits, necessary to represent the range of available ***brightness values***.
  - Data with an 8 bit radiometric resolution has 256 levels of brightness, while data with 12 bit radiometric resolution has 4,096 brightness levels.
- **Frame size:** the size of the recorded image frame is also an important property. It is described by the number of pixels across the frame or ***swath***, or in terms of the numbers of kilometres covered by the recorded scene.

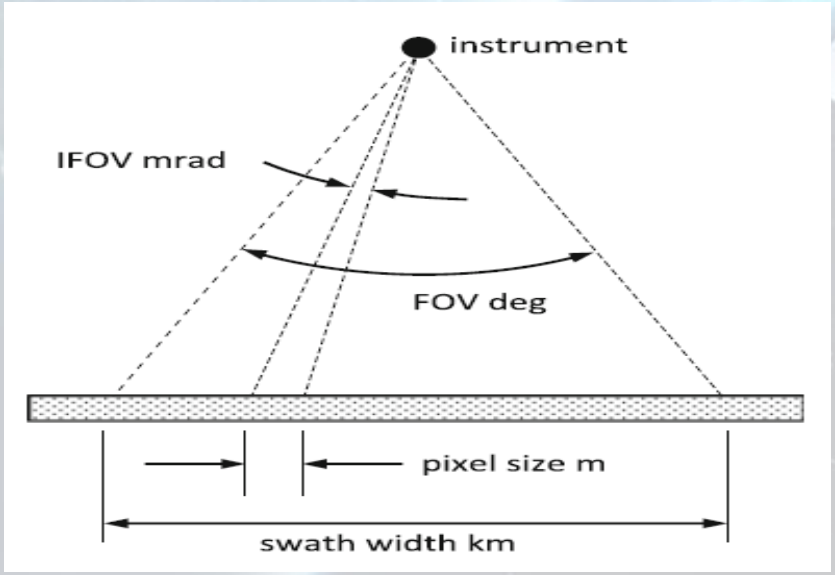


$N = 2^m$

m	N
8	256
10	1024
12	4096
16	65536

Primary Data Characteristics

- Image properties like ***pixel size*** and ***frame size*** are related directly to the technical characteristics of the sensor that was used to record the data.
- The **instantaneous field of view (IFOV)** of the sensor is its finest angular resolution. When projected onto the surface of the earth at the operating altitude of the platform, it defines the smallest resolvable element in terms of equivalent ground meters, which is what we refer to as *pixel size*.
- The **field of view (FOV)** of the sensor is the angular extent of the view it has across the earth's surface. When that angle is projected onto the surface it defines the *swath width* in equivalent ground kilometers.



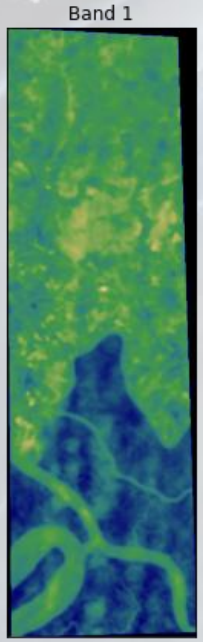
Example: Sentinel-2 Dataset

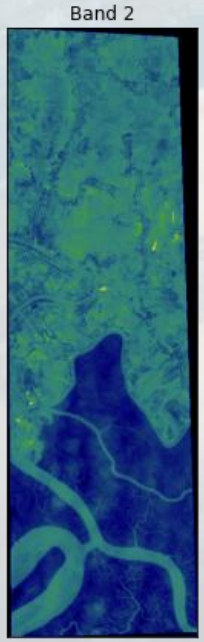
- Sentinel-2 (S2) is a wide-swath, high-resolution, multispectral imaging mission.
- The S2 Multispectral Instrument (MSI) samples 13 spectral bands ranging from 10 to 60-meter pixel size.
- There are **13** Sentinel-2 bands in total.
- Each band is **10, 20, or 60** meters in pixel size.
- Sentinel-2 consists of **2** satellites. First came Sentinel-2A which was launched in 2015. Next came Sentinel-2B in 2017.
- Two additional satellites (Sentinel-2C and 2D) were planned to launch in 2024.
  - Sentinel-2C was indeed launched in 2024.
  - Sentinel-2D is expected to launch in 2028.

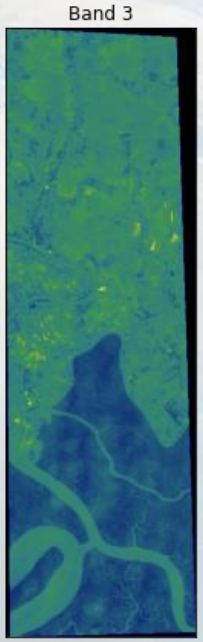
Example: Sentinel-2 Dataset

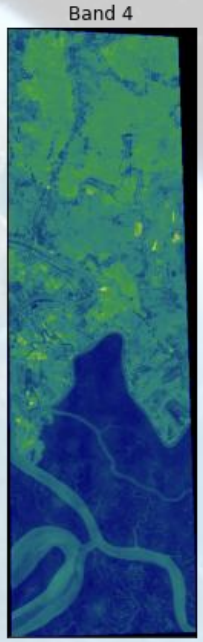
Band	Resolution	Central Wavelength	Description
B1	60 m	443 nm	Ultra Blue (Coastal and Aerosol)
B2	10 m	490 nm	Blue
B3	10 m	560 nm	Green
B4	10 m	665 nm	Red
B5	20 m	705 nm	Visible and Near Infrared (VNIR)
B6	20 m	740 nm	Visible and Near Infrared (VNIR)
B7	20 m	783 nm	Visible and Near Infrared (VNIR)
B8	10 m	842 nm	Visible and Near Infrared (VNIR)
B8a	20 m	865 nm	Visible and Near Infrared (VNIR)
B9	60 m	940 nm	Short Wave Infrared (SWIR)
B10	60 m	1375 nm	Short Wave Infrared (SWIR)
B11	20 m	1610 nm	Short Wave Infrared (SWIR)
B12	20 m	2190 nm	Short Wave Infrared (SWIR)

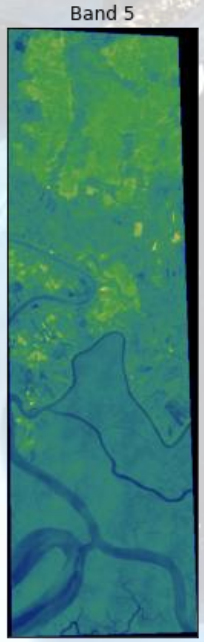
Example: Sentinel Bands Visualization

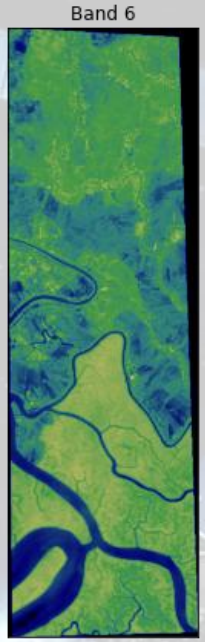


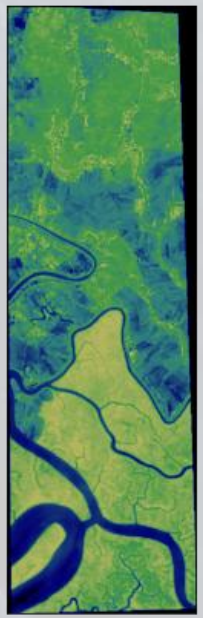


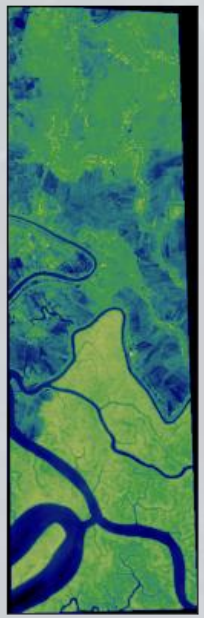


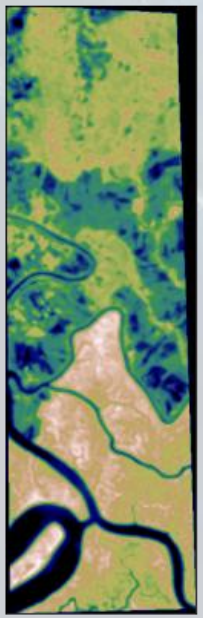


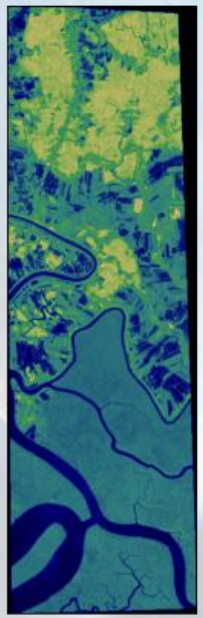


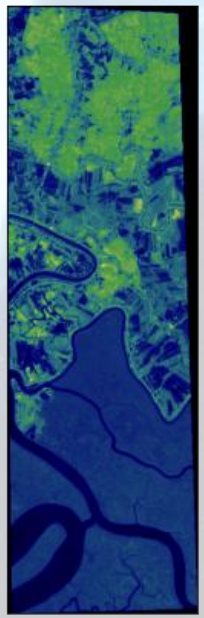


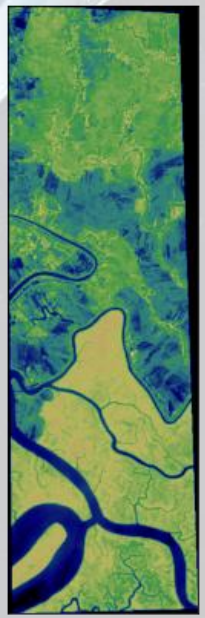












As viewed by human eyes (RGB Composite Image)



Example: Sentinel Band Combinations

- We use band combinations to better understand the features in imagery.
- The way we do this is by rearranging the available channels in creative ways.
- By using band combinations, we can extract specific information from an image.
  - For example, there are band combinations that highlight geologic, agricultural, or vegetation features in an image.

Example: Sentinel Band Combinations

Natural Color (B4, B3, B2):

- The natural color band combination uses the red (B4), green (B3), and blue (B2) channels. Its purpose is to display imagery the same way our eyes see the world.
- Just like how we see:
  - healthy vegetation is green.
  - urban features often appear white and grey.
  - water is a shade of dark blue depending on how clean it is.



Example: Sentinel Band Combinations

Color Infrared (B8, B4, B3):

- The color infrared band combination is meant to emphasize healthy and unhealthy vegetation.
- By using the near-infrared (B8) band, it's especially good at reflecting chlorophyll. This is why in a color infrared image, denser vegetation is red. But urban areas are white.



Example: Sentinel Band Combinations

Short-Wave Infrared (B12, B8A, B4):

- The short-wave infrared band combination uses SWIR (B12), NIR (B8A), and red (B4).
- This composite shows vegetation in various shades of green.
- In general, darker shades of green indicate denser vegetation. But brown is indicative of bare soil and built-up areas.



Example: Sentinel Band Combinations

Agriculture (B11, B8, B2)

- The agriculture band combination uses SWIR-1 (B11), near-infrared (B8), and blue (B2).
- It's mostly used to monitor the health of crops because of how it uses short-wave and near-infrared.
- Both these bands are particularly good at highlighting dense vegetation that appears as dark green.



Example: Sentinel Band Combinations

Geology (B12, B11, B2)

- The geology band combination is a neat application for finding geological features.
- This includes faults (an extended break in a rock formation), lithology (the study of the general physical characteristics of rocks), and geological formations.
- By leveraging the SWIR-2 (B12), SWIR-1 (B11), and blue (B2) bands, geologists tend to use this Sentinel band combination for their analysis.



Example: Sentinel Band Combinations

Bathymetric (B4, B3, B1)

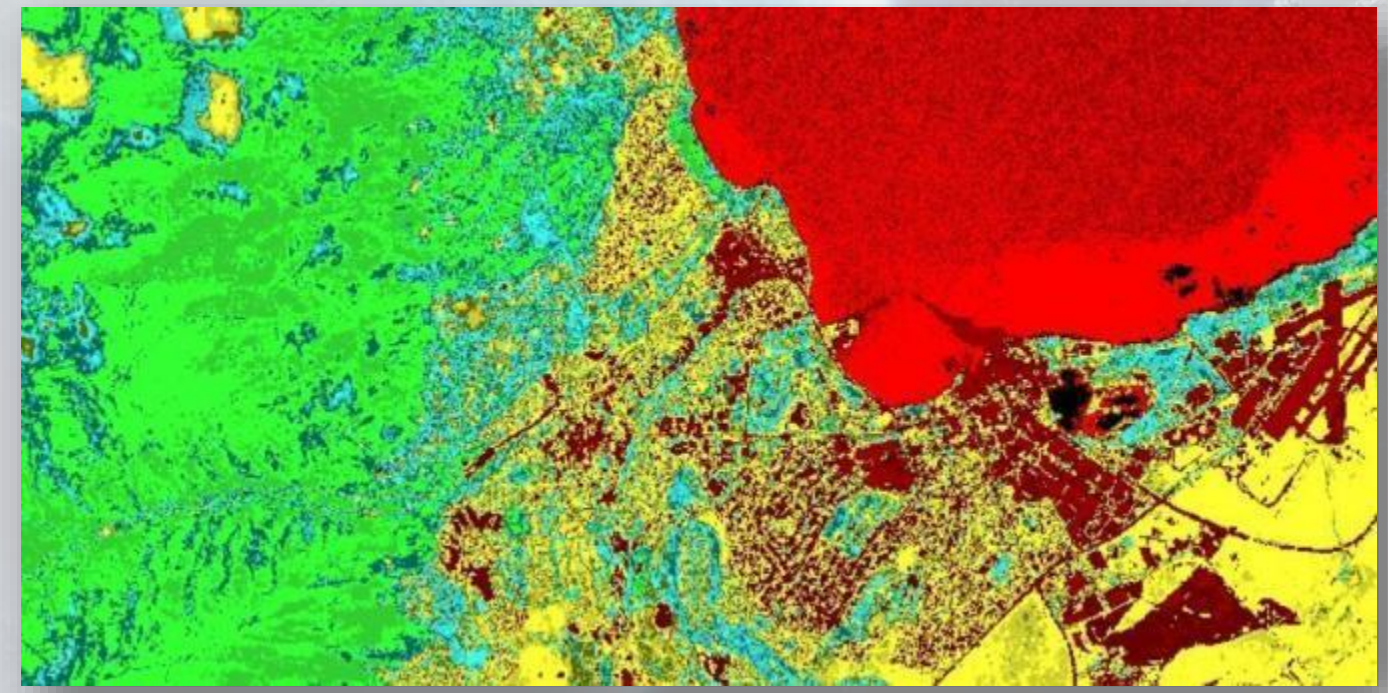
- As the name implies, the bathymetric band combination is good for coastal studies.
- The bathymetric band combination uses the red (B4), green (B3), and coastal band (B1). Using the coastal aerosol band is good for estimating suspended sediment in the water.



Example: Sentinel Band Combinations

Vegetation Index (B8-B4)/(B8+B4)

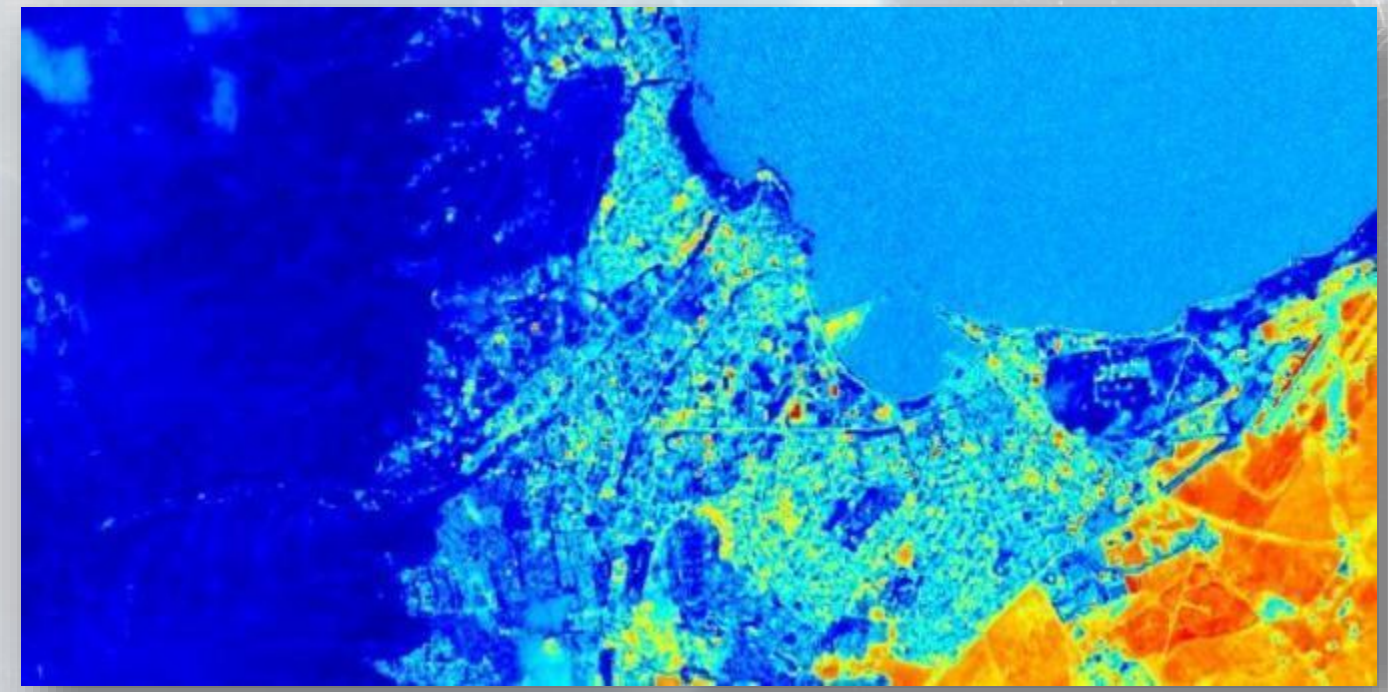
- Because near-infrared B8 (which vegetation strongly reflects) and red light B4 (which vegetation absorbs), the vegetation index is good for quantifying the amount of vegetation.
- The formula for the normalized difference vegetation index is (B8-B4)/(B8+B4).
- While high values suggest dense canopy (the uppermost branches of the trees in a forest), low or negative values indicate urban and water features.



Example: Sentinel Band Combinations

Moisture Index (B8A-B11)/(B8A+B11)

- The moisture index is ideal for finding water stress in plants.
- It uses the short-wave and near-infrared to generate an index of moisture content.
- In general, wetter vegetation has higher values. But lower moisture index values suggest plants are under stress from insufficient moisture.



Aerial Imagery vs Satellite Imagery

- Both offer a view of the ground and its features, objects from above:

  “bird’s eye view ”.

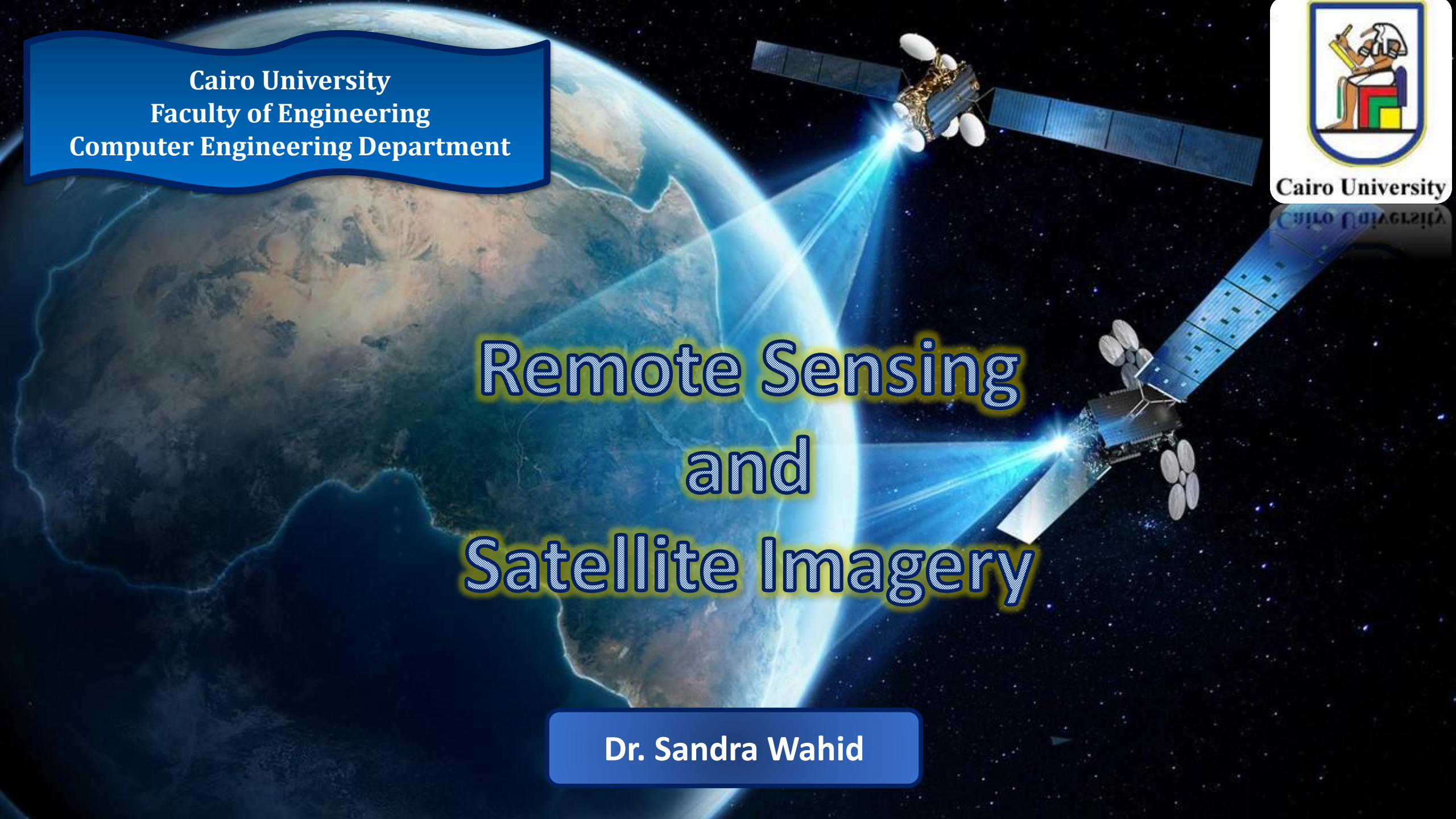


- The term **aerial imagery** refers to all imagery taken from an **airborne** craft which can include drones, balloons or airplanes.
- The term **satellite imagery** refers to all imagery taken by **artificial satellites**.

- While aerial photography implies images taken of the visible spectrum, **sensors** to measure bands within the nonvisible spectrum (e.g., ultraviolet, infrared, near-infrared) can also be fixed to aerial sources.


Thank You

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Computer Engineering Department



Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Types of Image Data Errors

Radiometric Errors

Errors in the measured brightness values of the pixels.

Geometric Errors

Errors between the actual image acquisition (sensor) coordinate data and ideal image coordinate data.

Note the following terminology:

BV: Brightness Value *or* **DN**: Digital Number

Sources of Radiometric Distortion

- Two broad types of radiometric distortion:
  - The **distribution of brightness** over an image in a given band can be different from that in the ground scene.
  - The **relative brightness** of a single pixel from band to band can be distorted.

- Both types can result from the presence of the **atmosphere** as a transmission medium through which radiation must travel from its source to the sensors, and can also be the result of **instrumentation effects**.

Correcting Instrumentation Errors

- Common remote sensing radiometric errors caused by instrumentation effects:
  - Random bad pixels (shot noise)
  - Line drop-outs
  - Line striping

Random Bad Pixels (Shot Noise)

- Sometimes an individual detector does not record spectral data for an **individual pixel**.
  - When this occurs **randomly**, it is called a **bad pixel**.
  - When there are **numerous random** bad pixels found within the scene, it is called **shot noise** because it appears that the image was shot by a shotgun.
- Normally these bad pixels contain values of **0 or 255** (in 8-bit data) in one or more of the bands.
- Shot noise is identified and repaired using the following methodology:
  - First locate each bad pixel in the band k dataset: a simple **thresholding** algorithm makes a pass through the dataset and flags any pixel ($BV_{i,j,k}$) having a brightness value of zero (while the surrounding pixels are not 0).
  - Once identified, it is then possible to evaluate the eight pixels surrounding the flagged pixel:

	col_{j-1}	col_j	col_{j+1}
row_{i-1}	BV_1	BV_2	BV_3
row_i	BV_8	BV_{ijk}	BV_4
row_{i+1}	BV_7	BV_6	BV_5

The mean of the eight surrounding pixels is computed and the value substitutes for $BV_{i,j,k}$ in the corrected image:

$$BV_{i,j,k} = \text{Int}\left[\left(\sum_{i=1}^{8} BV_i\right)/8\right]$$



15 17 16 14 16 19
13 [ ] 16 17 [ ] 14
16 15 14 17 16 15

After Correction:

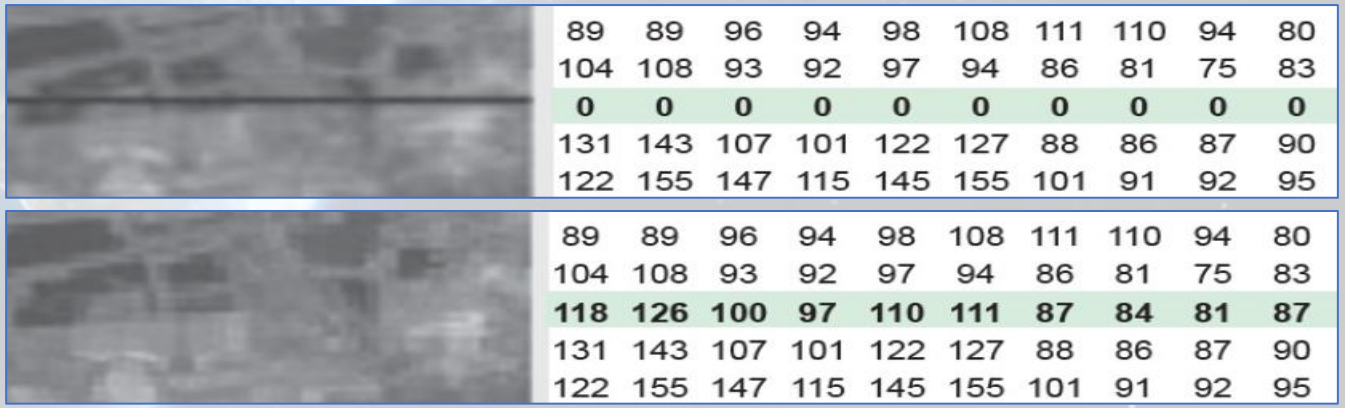
15 16

Line Drop-Outs

- If a detector in a linear array fails to function, this can result in an **entire line of data** with no spectral information.
- The bad line is commonly called a line drop-out and contains brightness values equal to zero.

- Methodology:
  - First locate each bad line in the image dataset: calculate the average ***BV*** per scan line for the entire scene. The average ***BV*** for each scan line is then compared with the scene average. Any scan line deviating from the average by more than a designated **threshold** value is identified as defective.
  - Once identified, it is then possible to collect the value for a pixel in the **preceding** line ($BV_{i-1,j,k}$) and the value for a pixel in the **succeeding** line ($BV_{i+1,j,k}$) and assign the output pixel ($BV_{i-1,j,k}$) in the drop-out line the **average** of these two brightness values:

$$BV_{i,j,k} = \text{Int}\left(\frac{BV_{i-1,j,k} + BV_{i+1,j,k}}{2}\right)$$



89	89	96	94	98	108	111	110	94	80
104	108	93	92	97	94	86	81	75	83
0	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
131	143	107	101	122	127	88	86	87	90
122	155	147	115	145	155	101	91	92	95

89	89	96	94	98	108	111	110	94	80
104	108	93	92	97	94	86	81	75	83
118	**126**	**100**	**97**	**110**	**111**	**87**	**84**	**81**	**87**
131	143	107	101	122	127	88	86	87	90
122	155	147	115	145	155	101	91	92	95

Line Striping

- Sometimes a detector does not fail completely, but simply goes out of **radiometric adjustment**.
- For example, a detector might record spectral measurements that are almost uniformly 20 brightness values greater than the other detectors for the same band.
- The result would be an image with systematic, noticeable lines that are brighter than adjacent lines. This is referred to as ***n-line striping***. The correction is referred to as ***destriping.***
- The maladjusted line contains valuable information, but should be corrected to have approximately the same radiometric scale as the data collected by the properly calibrated detectors associated with the same band.
- To repair systematic n-line striping, it is first necessary to identify the miscalibrated scan lines in the scene.
- This is usually accomplished by computing a **histogram** of the values for each of the n detectors that collected data over the entire scene.

Line Striping

- If one detector's mean or median is significantly different from the others, it is probable that this detector is out of adjustment.
- Consequently, every line and pixel in the scene recorded by the maladjusted detector may require a bias (**additive or subtractive**) correction or a more severe gain (**multiplicative**) correction.
- This type of n-line striping correction
  - adjusts all the bad scan lines so that they have approximately the same radiometric scale as the correctly collected data.
  - improves the visual interpretability of the data (makes it look better).



Line Striping

- Another Method:
  - Correction of radiometric mismatches among the detectors can be carried out by adopting one detector as a **standard** and adjusting the brightness of all pixels recorded by each other detector so that their mean brightness and standard deviations match those of the standard detector.
  - That operation can be implemented by:

$$y = \frac{\sigma_d}{\sigma_i} x + m_d - \frac{\sigma_d}{\sigma_i} m_i$$

Why this accomplishes the goal??

- x is the original brightness for a pixel
- y is its new (destriped) value
- m_d and σ_d are the reference values of mean brightness and standard deviation for the standard detector.
- m_i and σ_i are the values of mean brightness and standard deviation for the detector under consideration.

- Sometimes an *independent* reference mean and standard deviation is used.

Correcting Atmospheric Errors

- When an atmosphere is present there are two effects that must be taken into account: ***scattering*** and ***absorption*** by the particles.

- These lead to:
  - Fine detail in image data is obscured: *haze* in the imagery
  - The presence of the *absorption lines* superimposed by water vapour and other atmospheric constituents.

- Approximate techniques are available for atmospheric correction that depend directly on the measurements of the recorded image data.

- These are important when detailed data on the atmosphere, and particularly the water vapour content, are not available.

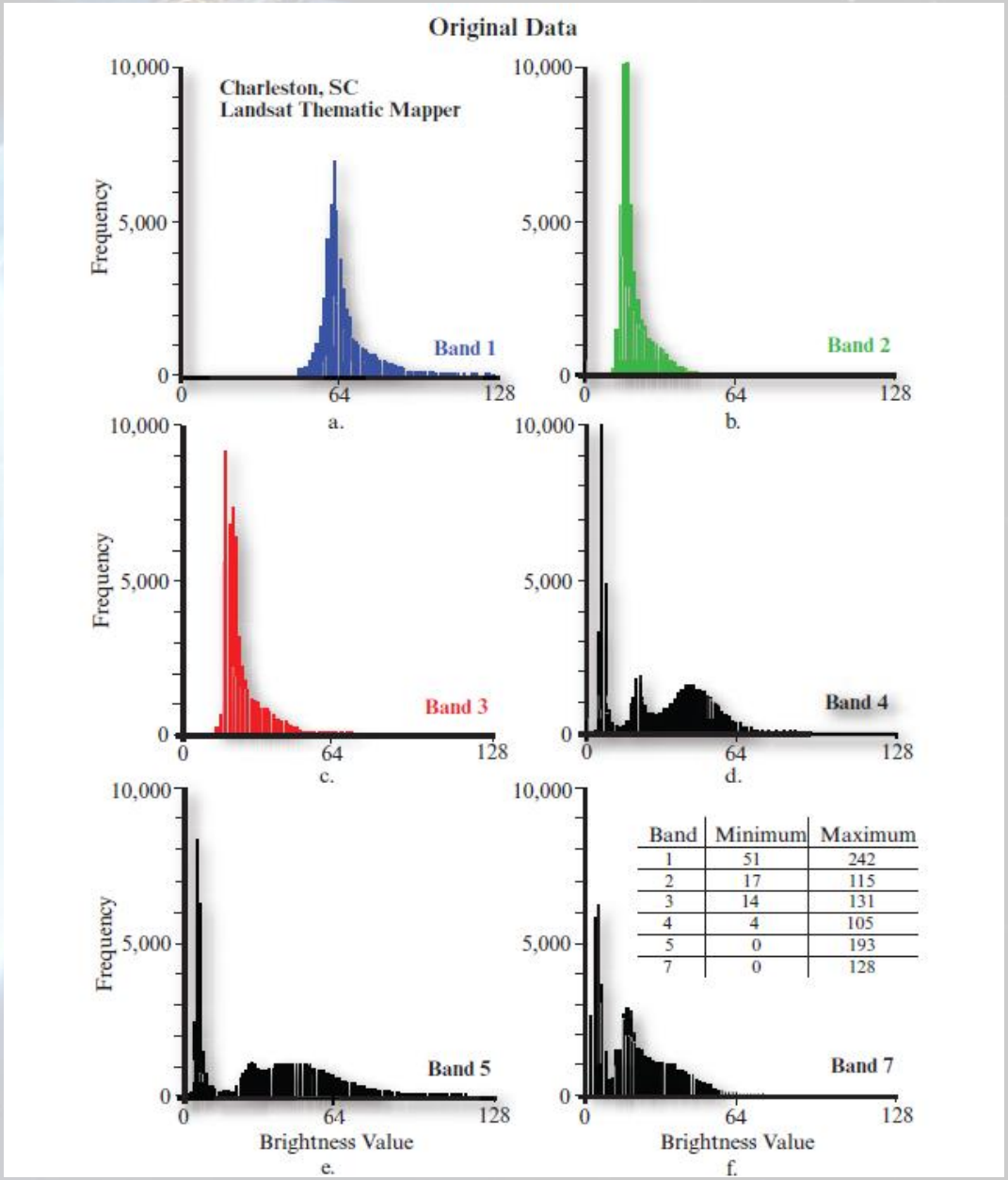
- The most common include:
  - Haze Removal by Dark Subtraction, also known as: Dark Object Subtraction

Haze Removal by Dark Subtraction

- It makes the assumption that each band of data for a given scene should contain some pixels at or close to **zero** brightness value but that atmospheric effects has added a **constant level** to each pixel in each band.

- Consequently, if histograms are taken of the bands (graphs of the number of pixels as a function of brightness value) the lowest significant occupied brightness value will be non-zero.

- The lowest occupied brightness value will be further from the origin for the *shorter wavelengths.*



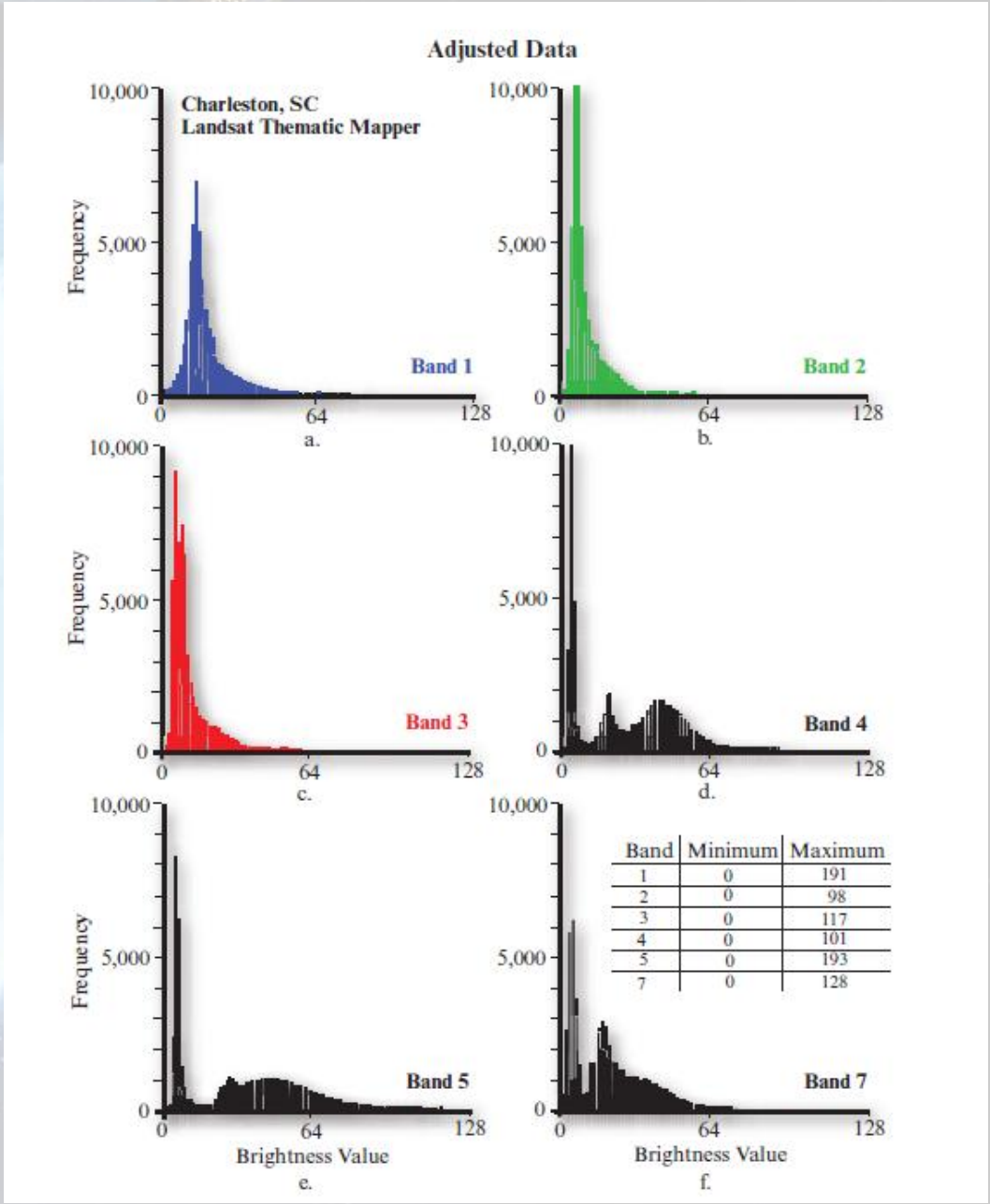
Haze Removal by Dark Subtraction

- If the histograms are shifted to the left so that zero values appear in the data, the effects of atmospheric scattering will be somewhat minimized.

- This simple algorithm is based on a **subtractive bias** established for each spectral band (the minimum *BV* in the histogram):

$$\text{output } BV_{i,j,k} = \text{input } BV_{i,j,k} - \text{bias}$$

- In the example the biases are chosen as follows:
  - Band1: bias=51
  - Band2: bias=17
  - Band3: bias=14
  - Band4: bias=4
  - Band5 and Band7: bias=0



Sources of Geometric Distortion

- Sources of geometric distortion include:
  - The rotation of the earth during image acquisition.
  - Variations in platform altitude and velocity.
  - The wide field of view of some sensors.
  - The curvature of the earth.
  - Sensor non-idealities.

- There are two techniques that can be used to correct the various types of geometric distortion present in digital image data:

Model the nature and magnitude of the **sources of distortion** and use the model to establish correction formulas.

This approach is effective when the types of distortion are well characterized, such as that caused by earth rotation.

Establish **mathematical relationships between the addresses of pixels in an image and the corresponding coordinates of those points on the ground** (via a map)→ **Image-to-Map Registration/Rectification**

This approach is the **most commonly used** and is independent of the analyst's knowledge of the source and type of distortion.

Use of Mapping Functions for Image Correction

- An assumption made in this procedure is the availability of a map of the region covered by the image, which is correct geometrically.
- We then define two Cartesian coordinate systems:
  - one describes the location of points in the map ***(x,y)***
  - the other defines the location of pixels in the image ***(u,v) or (x',y')***
- Suppose that the two coordinate systems can be related via a pair of mapping functions:

$$u = f(x, y)$$
$$v = g(x, y)$$

- First, we define a grid over the map to act as the grid of pixel centers for the corrected image, referred to as the ***display grid***
- We then move over the display grid pixel center by pixel center and use the mapping functions above to find the pixel in the image corresponding to each display grid position.
- Those pixels are then placed on the display grid.
- At the conclusion of the process, we have a geometrically correct image built up on the display grid using the original image as a source of pixels.
- <u>Problems:</u>

1. We do not know the explicit form of the ***mapping functions***.
2. Even if we did, for a given display grid location they may not point exactly to a pixel in the image. In such a case some form of ***interpolation*** will be required.

Mapping Polynomials and the Use of Ground Control Points

- Since explicit forms for the mapping functions are not known

→they are usually approximated by **polynomials** (first, second or third degree).

Sometimes orders higher than three are used but care must be taken to avoid the introduction of errors worse than those to be corrected.

- In the case of second degree:

$$u = a_0 + a_1x + a_2y + a_3xy + a_4x^2 + a_5y^2$$

$$v = b_0 + b_1x + b_2y + b_3xy + b_4x^2 + b_5y^2$$

x/u (column) and *y/v* (row) coordinates

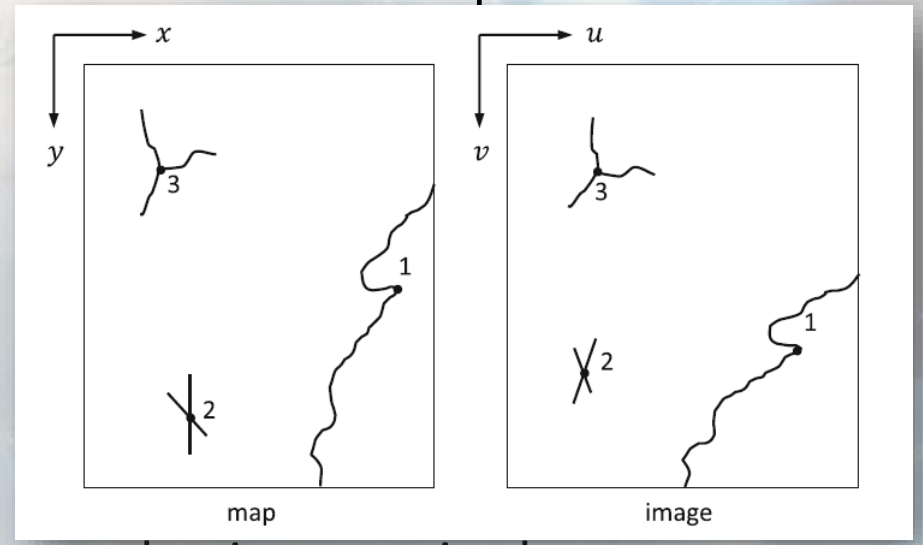
Known as: **Output-to-Input (Inverse) Mapping**
(u,v)→ original ***input*** image to be rectified
(x,y)→ ***output***-rectified image or map

- If the coefficients *ai* and *bi* were known then the mapping polynomials could be used to relate any point in the map to its corresponding point in the image.

- Coefficients values can be estimated by identifying sets of features on the map that can also be identified on the image. Those features, referred to as **ground control points (GCPs)** or just control points (CPs).

Mapping Polynomials and the Use of Ground Control Points

- These could be road intersections, street corners, airport runway intersections, sharp bends in rivers, coastline features, … etc.
- Enough are chosen, as pairs on the map and image so that the coefficients can be estimated by **substituting** the coordinates of the control points into the mapping polynomials to yield sets of equations in *ai* and *bi*.



- The minimum number of control points required:
  - for first order polynomial: 3
  - for second order polynomial: 6
  - for third order polynomial: 10
- In practice, significantly **more than those minimums** are chosen and the coefficients are evaluated using **least squares regression/estimation**.

→To avoid excessive influence on the estimated polynomial coefficients if any control points contain significant positional errors either on the map or in the image.

Problem 1 Solved

Ground Control Points

- Most image-to-map rectification still relies heavily on **human selection** of GCPs.
- A general rule is that there should be a distribution of control points around the **edges** of the image to be corrected, with a **scattering** of points over the body of the image →that is necessary to ensure that the mapping polynomials are well-behaved over the scene.
- After computing equations' coefficients, it is important to determine how well the coefficients account for the geometric distortion in the input image.

The method often used is root-mean-square error (RMS_{error})

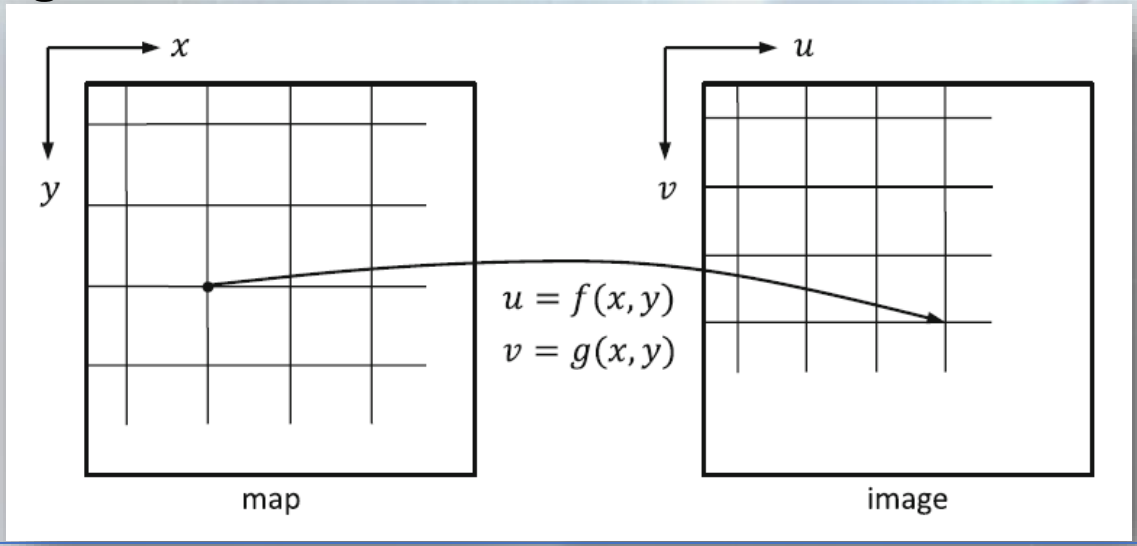
- Let x_{orig} and y_{orig} be the column and row coordinates of GCP in the original input image.
- Let x and y be the column and row coordinates of GCP in the reference map.
- Let x' and y' be the column and row coordinates of GCP from the equation.

→ Ideally $x' = x_{orig}$ and $y' = y_{orig}$

- For each GCP, compute: $$RMS_{error} = \sqrt{(x' - x_{orig})^2 + (y' - y_{orig})^2}$$
- By computing RMS_{error} for all GCPs, it is possible to:
  - 1) see which GCPs exhibit the greatest error 2) sum all the RMS_{error}
- The user specifies a threshold of acceptable total RMS_{error}. If the threshold is exceeded:
  - 1) delete the GCP that has the greatest RMS_{error} 2) recompute the coefficients 3) recompute the RMS_{error} for all points
  - This process continues until the total RMS_{error} is less than the threshold specified or too few points remain to perform a least-squares regression to compute the coefficients.

Building a Geometrically Correct Image

- Referred to as: **rectified image.**
- Having specified the mapping polynomials completely by the use of ground control points →the next step is to find points in the image that correspond to each location in the display grid.
- The spacing of that grid is chosen according to the **pixel size required** in the corrected image and need not be the same as that of the original, geometrically distorted version.
- For the moment suppose that the points located in the image correspond exactly to image pixel centers, even though that rarely happens in practice.
- Then those pixels are **simply transferred** to the appropriate locations on the display grid to build up the rectified image.



Using mapping functions to locate points in the image corresponding to particular display grid (map) positions

Resampling and the Need for Interpolation

- As expected, grid centers from the map-defined display grid will not usually project to exact pixel center locations in the image.

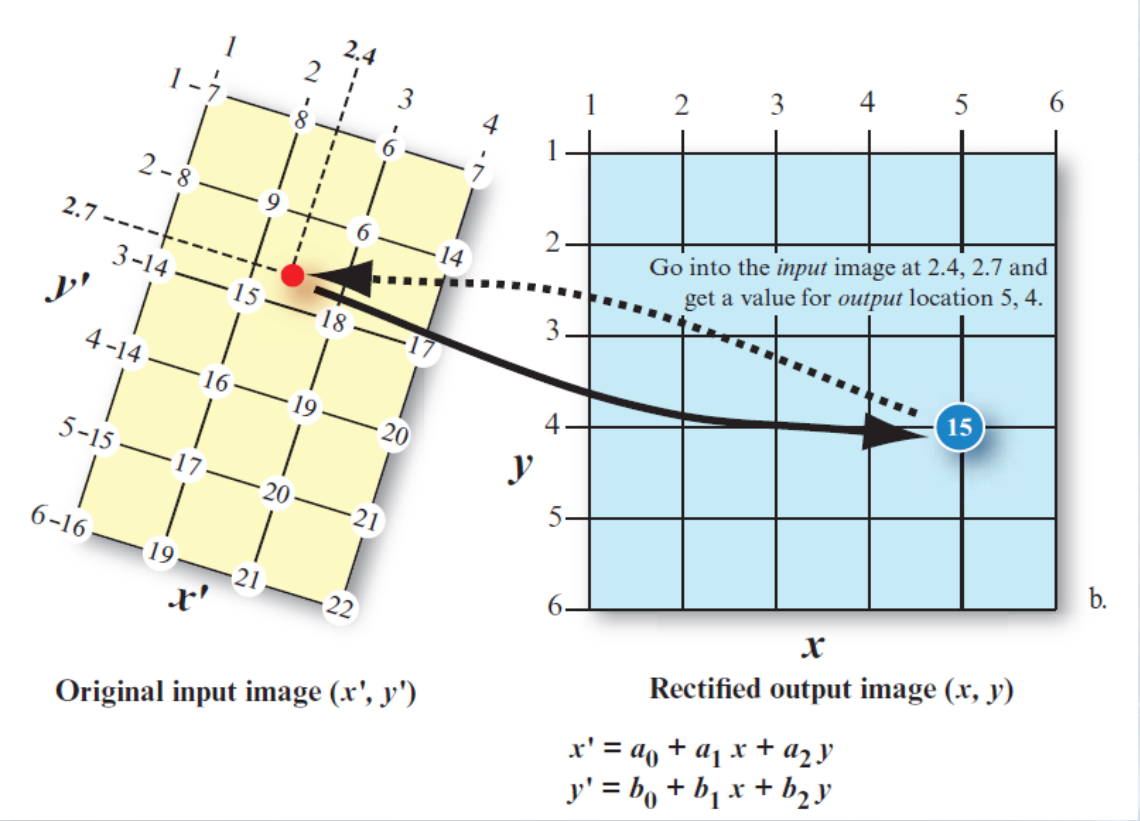
- Some decision now has to be made about what pixel brightness value should be chosen for placement on the new grid.

- Three principal techniques are used for this purpose:
  - Nearest Neighbor Resampling
  - Bilinear Interpolation
  - Cubic Convolution Interpolation

Problem 2 Solved

Nearest Neighbor Resampling

- Simply chooses the actual pixel that has **its center nearest to the point located in the image**.
- That pixel value is transferred to the corresponding display grid location.
- This is the preferred technique if the new image is to be classified (e.g.: distinguishing one type of vegetation from another) because it consists only of original pixel brightness values, simply rearranged in position to give the correct image geometry.
- The method is only acceptable when the new and old pixel sizes and spacings are not too different.



Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x′, y′ to the Sample Point, D
2, 2	9	$D = \sqrt{(2.4-2)^2 + (2.7-2)^2} = 0.806$
3, 2	6	$D = \sqrt{(2.4-3)^2 + (2.7-2)^2} = 0.921$
2, 3	15	$D = \sqrt{(2.4-2)^2 + (2.7-3)^2} = 0.500$
3, 3	18	$D = \sqrt{(2.4-3)^2 + (2.7-3)^2} = 0.670$

Bilinear Interpolation

- First-order or bilinear interpolation assigns output pixel values by interpolating brightness values in two orthogonal directions in the input image.
- It basically fits a plane to the **four pixel values** nearest to the desired position *(x',y')* in the input image and then computes a new brightness value based on the weighted distances to these points.
- The closer a pixel is to the desired location, the more weight it will have in the final computation of the average.
- The weighted average of the new brightness value ($\boldsymbol{BV_{wt}}$) is computed using the equation:

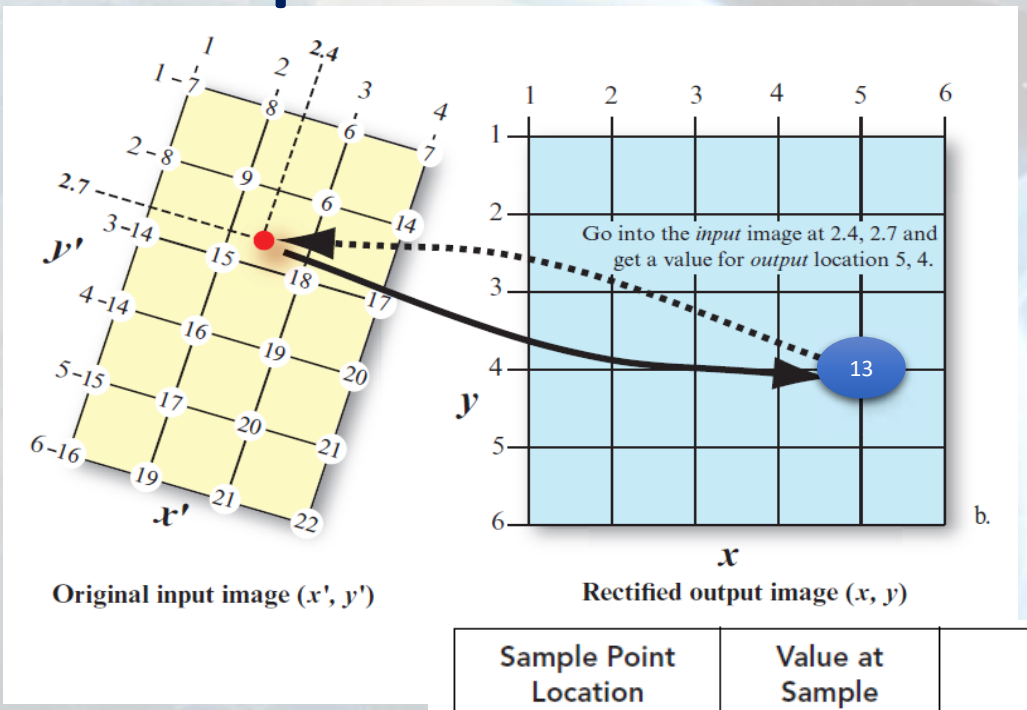
$$\text{Bilinear}_{BV_{wt}} = \frac{\sum_{k=1}^{4} \frac{Z_k}{D_k^2}}{\sum_{k=1}^{4} \frac{1}{D_k^2}},$$

Where:

- Z_k are the surrounding 4 data point values.
- D_k^2 are the distances squared from the point in question (x',y') to these data points.

Bilinear Interpolation

- Example:



Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x′, y′ to the Sample Point, D	D_k^2	$\frac{Z}{D_k^2}$	$\frac{1}{D_k^2}$
2, 2	9	$D = \sqrt{(2.4-2)^2 + (2.7-2)^2} = 0.806$	0.65	13.85	1.539
3, 2	6	$D = \sqrt{(2.4-3)^2 + (2.7-2)^2} = 0.921$	0.85	7.06	1.176
2, 3	15	$D = \sqrt{(2.4-2)^2 + (2.7-3)^2} = 0.500$	0.25	60.00	4.000
3, 3	18	$D = \sqrt{(2.4-3)^2 + (2.7-3)^2} = 0.670$	0.45	40.00	2.222
				Σ 120.91	Σ 8.937
				BV_{wt} = 120.91/8.937 =13.53	

Truncate to 13

Cubic Convolution Interpolation

- Assigns values to output pixels in the same manner as bilinear interpolation, except that **sixteen input pixels** surrounding the location of the desired pixel are used.
- The weighted average of the new brightness value ($\boldsymbol{BV_{wt}}$) is computed using the equation:

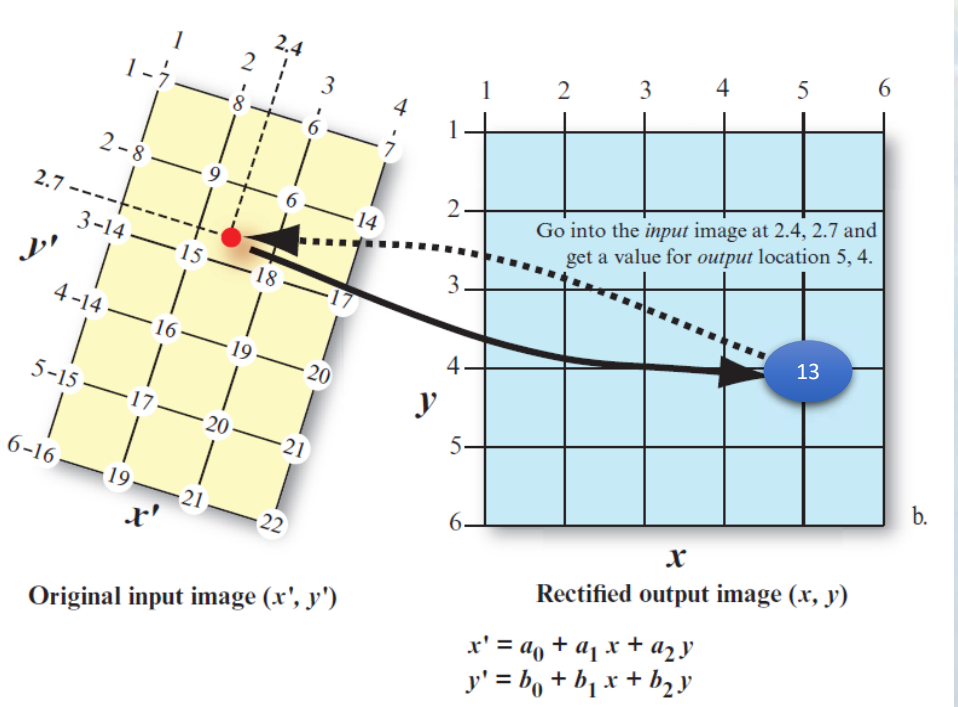
$$\text{Cubic Convolution}_{BV_{wt}} = \frac{\sum_{k=1}^{16} \frac{Z_k}{D_k^2}}{\sum_{k=1}^{16} \frac{1}{D_k^2}},$$

Where:

- Z_k are the surrounding 16 data point values.
- D_k^2 are the distances squared from the point in question (x',y') to these data points.

Cubic Convolution Interpolation

- Example:



$$x' = a_0 + a_1 x + a_2 y$$
$$y' = b_0 + b_1 x + b_2 y$$

Sample Point Location (column, row)	Value at Sample Point, Z	Distance from x', y' to the Sample Point, D	D_k^2	$\frac{Z}{D_k^2}$	$\frac{1}{D_k^2}$
1, 1	7	$D = \sqrt{(2.4-1)^2+(2.7-1)^2} = 2.202$	4.85	1.443	0.206
2, 1	8	$D = \sqrt{(2.4-2)^2+(2.7-1)^2} = 1.746$	3.05	2.623	0.328
3, 1	6	$D = \sqrt{(2.4-3)^2+(2.7-1)^2} = 1.80$	3.24	1.852	0.309
4, 1	7	$D = \sqrt{(2.4-4)^2+(2.7-1)^2} = 2.335$	5.45	1.284	0.183
1, 2	8	$D = \sqrt{(2.4-1)^2+(2.7-2)^2} = 1.565$	2.45	3.265	0.408
2, 2	9	$D = \sqrt{(2.4-2)^2+(2.7-2)^2} = 0.806$	0.65	13.85	1.539
3, 2	6	$D = \sqrt{(2.4-3)^2+(2.7-2)^2} = 0.921$	0.85	7.06	1.176
4, 2	14	$D = \sqrt{(2.4-4)^2+(2.7-2)^2} = 1.746$	3.05	4.59	0.328
1, 3	14	$D = \sqrt{(2.4-1)^2+(2.7-3)^2} = 1.432$	2.05	6.829	0.488
2, 3	15	$D = \sqrt{(2.4-2)^2+(2.7-3)^2} = 0.500$	0.25	60.00	4.000
3, 3	18	$D = \sqrt{(2.4-3)^2+(2.7-3)^2} = 0.670$	0.45	40.00	2.222
4, 3	17	$D = \sqrt{(2.4-4)^2+(2.7-3)^2} = 1.63$	2.65	6.415	0.377
1, 4	14	$D = \sqrt{(2.4-1)^2+(2.7-4)^2} = 1.911$	3.65	3.836	0.274
2, 4	16	$D = \sqrt{(2.4-2)^2+(2.7-4)^2} = 1.360$	1.85	8.649	0.541
3, 4	19	$D = \sqrt{(2.4-3)^2+(2.7-4)^2} = 1.432$	2.05	9.268	0.488
4, 4	20	$D = \sqrt{(2.4-4)^2+(2.7-4)^2} = 2.062$	4.25	4.706	0.235
				Σ 175.67	Σ 13.102
				BV_{wt} = 175.67 / 13.102	
				BV_{wt} = **13.41**	

Truncate to 13

Image-to-Image Registration

- Many applications in remote sensing (e.g. change detection) require two or more scenes of the same geographical region, acquired at different times, to be processed together.

- Two images can be registered to each other by registering each to a map coordinate base separately in the manner previously demonstrated.

- Alternatively, and particularly if georeferencing is not important, one image can be chosen as a **master, or reference**, to which the other, known as the **slave**, is registered.

- Same techniques as image-to-map registration are used.
- However, the coordinates are now the pixel coordinates in the master image rather than the map coordinates.

- A benefit in image-to-image registration is that only one registration step is required, by comparison to two if both are taken back to a map base.

Other Image Geometry Operations

- Intentional changes to image geometry can be performed including:
  - Image Rotation
  - Scale Changing and Zooming

Image Rotation:

To rotate an image in an anticlockwise sense by any specified angle ζ:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\zeta & \sin\zeta \\ -\sin\zeta & \cos\zeta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\zeta & -\sin\zeta \\ \sin\zeta & \cos\zeta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

Negative the angle

Other Image Geometry Operations

Image Scale Changing and Zooming:

- The scales of an image in both the vertical and horizontal directions can be altered by the transformation:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}$$

a and *b* are the desired scaling factors

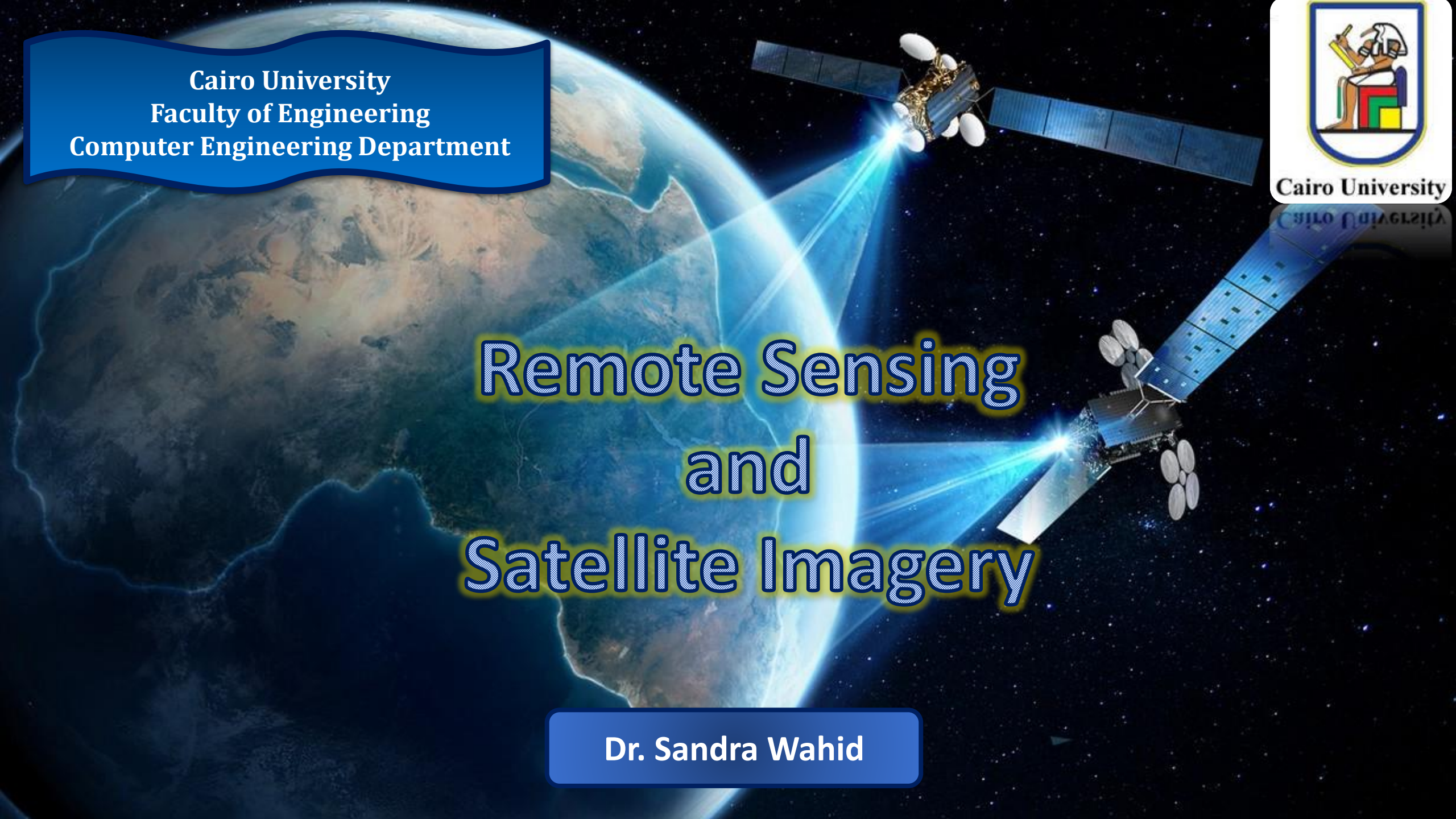
- To resample the scaled image onto the display grid we use the inverse operation to locate pixel positions in the original image corresponding to each display grid position:

$$\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 1/a & 0 \\ 0 & 1/b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- Interpolation is used to establish the actual pixel brightness values to use, since *u* and *v* will not normally fall on exact pixel locations.
- Frequently *a=b* so that the image is simply magnified called **zooming**.
- If the *nearest neighbor interpolation* procedure is used in the resampling process the zoom implemented is said to occur by ***pixel replication*** and the image will look progressively *blocky* for larger zoom factors.
- If *cubic convolution interpolation* is used there will be a *change in magnification* but the image will *not take on the blocky* appearance. Often this process is called ***interpolative zoom***.


Thank You

Cairo University
Faculty of Engineering
Computer Engineering Department

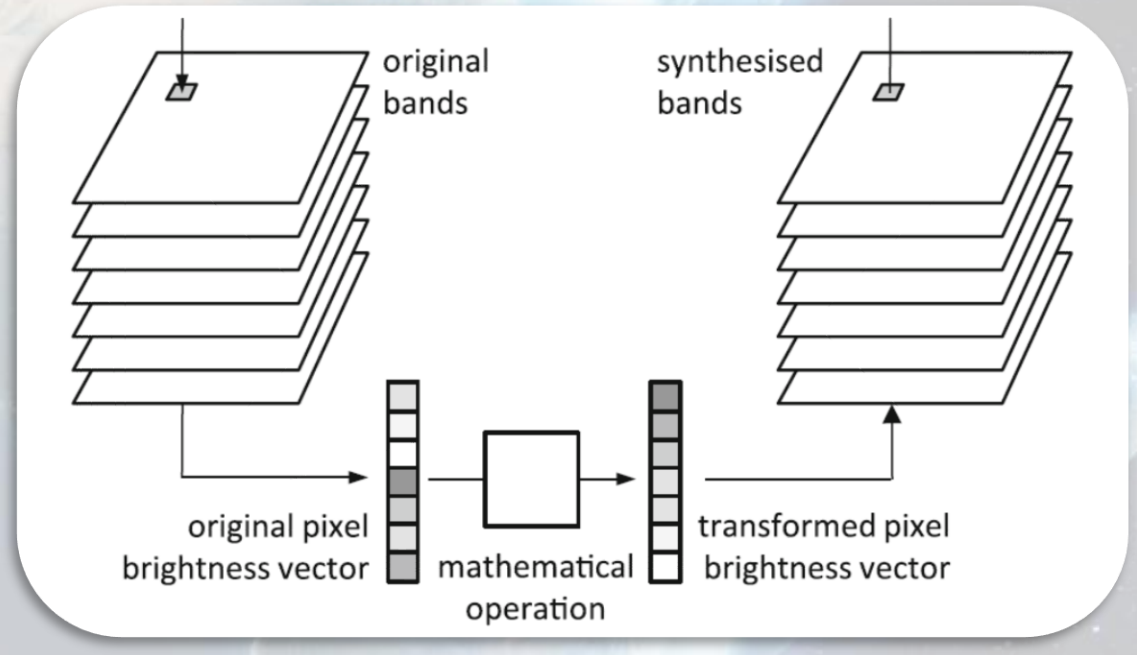


Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Introduction

- Because of the multispectral or vector character of remote sensing image data, we can devise transformations that generate new pixel vector descriptions from the old, and thereby **create synthetic image components, or bands**.



- They represent an alternative description of the data. The new components are related to the old brightness values of the pixels in the original set of spectral bands via a **mathematical operation**, which is usually linear.

Introduction

- Why perform such transformations?

May make evident features that are not easily seen in the original data by **devising pixel descriptors matched to particular applications**

- Vegetation indices are a good example used for indicating the degree of vegetation.

It might be possible to preserve most of the essential information content of the original image using a **reduced number of the transformed dimensions**

- This has significance for compressed transmission and storage of data.

Image Arithmetic and Vegetation Indices

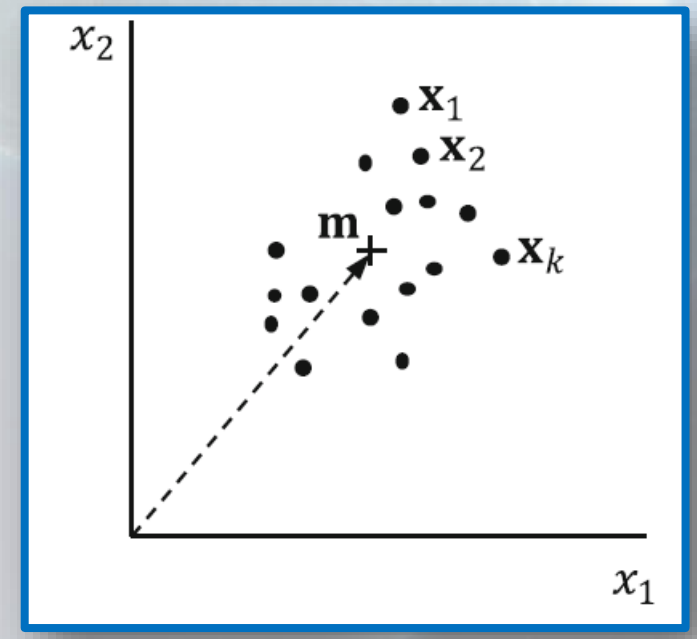
- **Addition, subtraction and division** (ratios) of the brightness values of two or more bands are the most common. While product operations are rarely encountered.
  - Differences can be used to highlight regions of change between two images of the same area. That requires the images to be first registered.
  - The resultant difference image must be scaled to remove negative brightness values; normally that is done so that regions of no change appear mid-grey, while changes show as brighter or duller than mid-grey according to the sign of the difference.

- More sophisticated vegetation indices can be created using **composite band arithmetic** operations.
  - Such as the widely used: “Normalised Difference Vegetation Index (NDVI)”

$$NDVI = \frac{\phi_{\text{nir}} - \phi_{\text{red}}}{\phi_{\text{nir}} + \phi_{\text{red}}}$$

The Principal Components Transformation

- We can represent pixels in a **vector space:** called **“a spectral space”**
  - Each pixel is a vector.
  - The vector dimension = number of spectral components associated with each pixel.
- A particular pixel plots as a point in such a space with coordinates that correspond to its brightness values in the spectral components.
- Example: a two dimensional spectral space showing individual pixel vectors and their mean vector m.



The Mean Vector and The Covariance Matrix

- The mean position of the pixels in the space is defined by the expected (or average) value of the pixel vector **x**:

$$\mathbf{m} = \mathcal{E}\{\mathbf{x}\} = \frac{1}{K}\sum_{k=1}^{K}\mathbf{x}_k$$

where $\mathbf{m}$ is the mean pixel vector and the $\mathbf{x}_k$ are the individual vectors of total number K; $\mathcal{E}$ is the expectation operator.

- To describe pixels ***scatter or spread*** → use the **covariance matrix**:

$$\mathbf{C}_x = \mathcal{E}\left\{(\mathbf{x}-\mathbf{m})(\mathbf{x}-\mathbf{m})^{\mathrm{T}}\right\}$$

$$\mathbf{C}_x = \frac{1}{K-1}\sum_{k=1}^{K}(\mathbf{x}_k-\mathbf{m})(\mathbf{x}_k-\mathbf{m})^{\mathrm{T}}$$

sample covariance

→ Symmetric Matrix

Cov(a,a)=Var(a)
Cov(a,b)=Cov(b,a)

- To examine the properties of the covariance matrix, another matrix is defined called the **correlation matrix $\mathbf{R_x}$**:

$$r_{ij} = c_{ij}/\sqrt{c_{ii}c_{jj}} = r_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\sigma_Y}$$

→ Symmetric Matrix

It explicitly describes the degree of **correlation** between the data dimensions/spectral measurements/coordinates

Example

Set 1:

$\mathbf{x}$	$\mathbf{x}-\mathbf{m}$	$(\mathbf{x}-\mathbf{m})(\mathbf{x}-\mathbf{m})^T$
$\begin{bmatrix}1\\2\end{bmatrix}$	$\begin{bmatrix}-2.00\\-0.33\end{bmatrix}$	$\begin{bmatrix}4.00 & 0.66\\0.66 & 0.11\end{bmatrix}$
$\begin{bmatrix}2\\1\end{bmatrix}$	$\begin{bmatrix}-1.00\\-1.33\end{bmatrix}$	$\begin{bmatrix}1.00 & 1.33\\1.33 & 1.77\end{bmatrix}$
$\begin{bmatrix}4\\1\end{bmatrix}$	$\begin{bmatrix}1.00\\-1.33\end{bmatrix}$	$\begin{bmatrix}1.00 & -1.33\\-1.33 & 1.77\end{bmatrix}$
$\begin{bmatrix}5\\2\end{bmatrix}$	$\begin{bmatrix}2.00\\-0.33\end{bmatrix}$	$\begin{bmatrix}4.00 & -0.66\\-0.66 & 0.11\end{bmatrix}$
$\begin{bmatrix}4\\4\end{bmatrix}$	$\begin{bmatrix}1.00\\1.67\end{bmatrix}$	$\begin{bmatrix}1.00 & 1.67\\1.67 & 2.79\end{bmatrix}$
$\begin{bmatrix}2\\4\end{bmatrix}$	$\begin{bmatrix}-1.00\\1.67\end{bmatrix}$	$\begin{bmatrix}1.00 & -1.67\\-1.67 & 2.79\end{bmatrix}$

from which $\mathbf{C}_x = \begin{bmatrix}2.40 & 0\\0 & 1.87\end{bmatrix}$ and $\mathbf{R}_x = \begin{bmatrix}1.00 & 0\\0 & 1.00\end{bmatrix}$

The mean vector for this data set is $\mathbf{m} = \begin{bmatrix}3.00\\2.33\end{bmatrix}$

The covariance matrix and consequently the correlation matrix, have **zero off diagonal entries**

→indicating that there is **no correlation** between the two spectral components of the data set

Set 2:

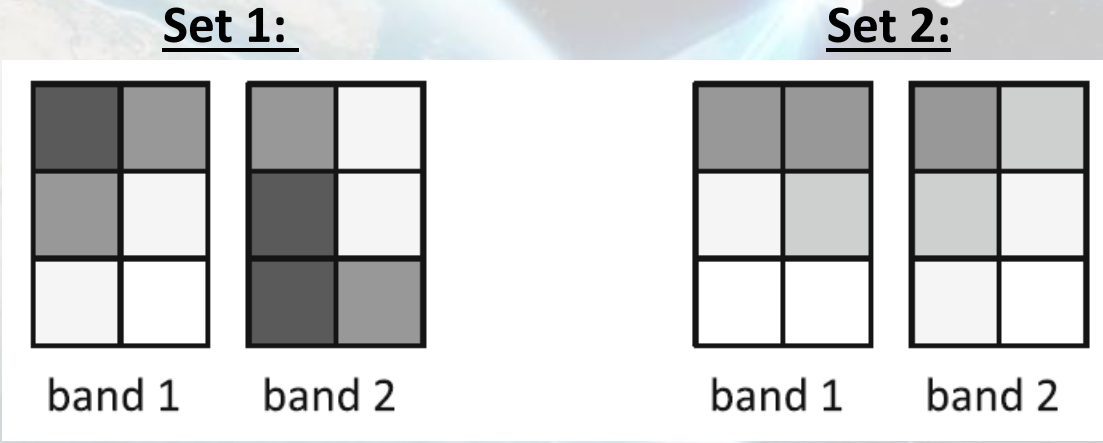
$$\mathbf{m} = \begin{bmatrix}3.50\\3.50\end{bmatrix} \quad \mathbf{C}_x = \begin{bmatrix}1.90 & 1.10\\1.10 & 1.10\end{bmatrix}$$

$$\mathbf{R}_x = \begin{bmatrix}1.00 & 0.76\\0.76 & 1.00\end{bmatrix}$$

Shows the two spectral components to be **76% correlated**

Correlation

- Showing what the images represented by the data in the previous example might look like:



- Set 1: the uncorrelated data, there is no near-correspondence in dark and light between the bands.
- Set 2: when one band is dark the other is generally dark, and vice versa
  - →For example: it would be possible to **predict** the pixel in one band knowing its value in the other band.
  - →There is redundancy.
- Different Correlations:



No relation

When x1 increase →x2 decrease

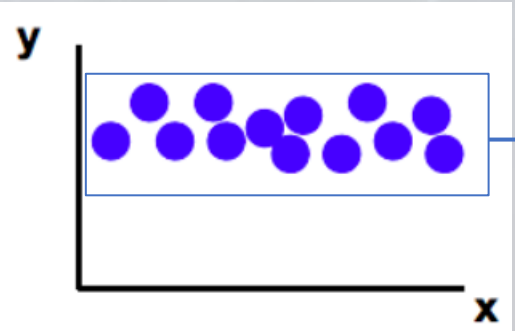
When x1 increase →x2 increase

Principal Component Analysis (PCA)

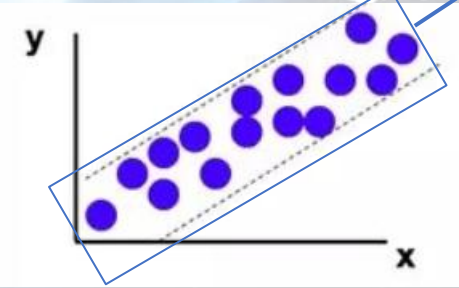
- PCA has proven to be of great value in the analysis of multispectral remotely sensed data.
- PCA is a technique that transforms the original remotely sensed dataset into a substantially smaller and easier to interpret set of uncorrelated variables that represents most of the information present in the original dataset.

Less Correlation→ more information.
More Correlation→ less information (contain redundant information).

Less Variance→ less information.
More Variance→ more information.



y small range (variance)
x large range (variance)
→x has more information
so it is good to be kept



small variance direction is a combination of x, y
So can rotate so that directions of variations are parallel to co-ordinate axis

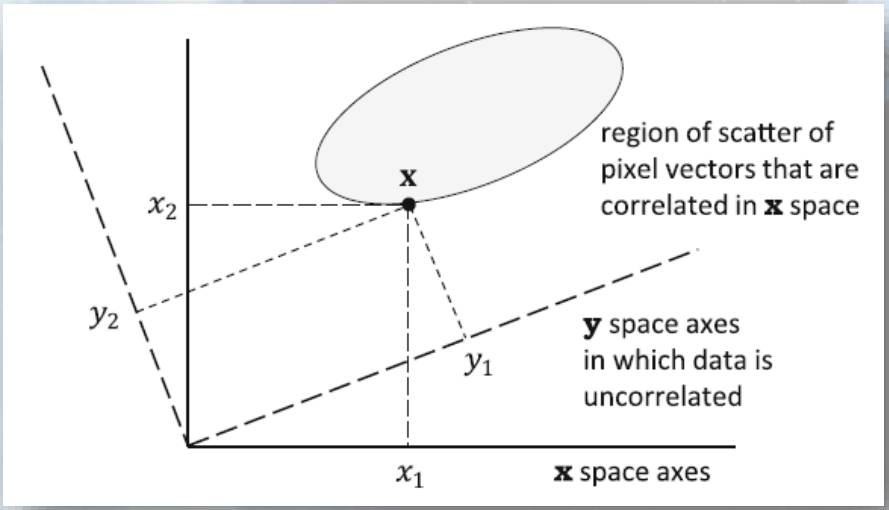
PCA is a technique for reducing the dimensionality of a dataset.

Principal Component Analysis (PCA)

- It is fundamental to the development of the principal components transformation to ask whether there is a ***new coordinate system*** in spectral space in which the data can be represented ***without correlation***

→ the **covariance matrix** in the new coordinates **is diagonal**.

- Consider the example of 2 dimensions:



- If the vectors describing the pixel points are represented as y in the new coordinates then we want to find a linear transformation G of the original x coordinates, such that:

$$\mathbf{y} = \mathbf{G}\mathbf{x} = \mathbf{D}^T\mathbf{x}$$

subject to the condition (constraint) that the covariance matrix of the pixel data in the y coordinates is diagonal

D is an orthonormal matrix (since it is a rotation matrix)

$\mathbf{D}^T=\mathbf{D}^{-1}$

→ columns and rows are unit vectors and orthogonal

Principal Component Analysis (PCA)

- In the y coordinate space the covariance matrix of the data is:

$$\mathbf{C}_y = \mathcal{E}\left\{(\mathbf{y}-\mathbf{m}_y)(\mathbf{y}-\mathbf{m}_y)^{\mathrm{T}}\right\}$$

$$\mathbf{m}_y = \mathcal{E}\{\mathbf{y}\} = \mathcal{E}\{\mathbf{D}^{\mathrm{T}}\mathbf{x}\} = \mathbf{D}^{\mathrm{T}}\mathcal{E}\{\mathbf{x}\} = \mathbf{D}^{\mathrm{T}}\mathbf{m}_x$$

Since D^T is a matrix of constants it can be taken outside the expectation operator

$$\mathbf{C}_y = \mathcal{E}\left\{(\mathbf{D}^{\mathrm{T}}\mathbf{x}-\mathbf{D}^{\mathrm{T}}\mathbf{m}_x)(\mathbf{D}^{\mathrm{T}}\mathbf{x}-\mathbf{D}^{\mathrm{T}}\mathbf{m}_x)^{\mathrm{T}}\right\}$$

$$\mathbf{C}_y = \mathbf{D}^{\mathrm{T}}\mathcal{E}\left\{(\mathbf{x}-\mathbf{m}_x)(\mathbf{x}-\mathbf{m}_x)^{\mathrm{T}}\right\}\mathbf{D}$$

$$\mathbf{C}_y = \mathbf{D}^{\mathrm{T}}\mathbf{C}_x\mathbf{D} = \mathbf{G}\mathbf{C}_x\mathbf{G}^{\mathrm{T}}$$

Recall: $AV = V\Lambda$
where:

- A is a matrix
- V is the Eigen Vectors matrix of A
- Λ is the Eigen Values matrix of A → is a diagonal matrix

So $V^{-1}AV = V^{-1}V\Lambda$

$\boldsymbol{V^{-1}AV = \Lambda}$

Eigenvectors matrix of a symmetric matrix is orthonormal

- **Cy: matrix of the eigenvalues of Cx**
- **D: matrix of the eigenvectors of Cx**: each column is an eigenvector
- Note: $D^T=D^{-1}$ (orthonormal matrix)

Our goal is achieved:
Cy is a diagonal matrix

Principal Component Analysis (PCA)

- In the y coordinate space the covariance matrix of the data is:

$$\mathbf{C}_y = \begin{bmatrix} \lambda_1 & 0 & & & \\ 0 & \lambda_2 & & & \\ & & \cdot & & \\ & & & \cdot & \\ & & & \lambda_{N-1} & 0 \\ & & & 0 & \lambda_N \end{bmatrix}$$

  - N is the dimensionality of the data (the number of spectral bands)
  - λ_n are the eigenvalues of Cx
  - The matrix is arranged diagonally such that $\lambda_1 > \lambda_2 \ldots > \lambda_N$
  - **Eigenvectors** of Cx are the **directions** of the axes where there is the most variance (most information) and that we call *Principal Components*.
  - **Eigenvalues** of Cx are the **variances** along each of the new coordinates in the y space.
  - In view of the rank ordering of the eigenvalues, the data will show greatest spread along the first y coordinate. Its second greatest dispersion will be along the second new coordinate, which corresponds to the second largest eigenvalue, and so on.
  - We can take only **L<N directions** corresponding to the largest L eigenvalues and ignore the rest.

Principal Component Analysis (PCA)

- To summarize the steps:

1. Generate the covariance matrix of the data in the coordinates of the original spectral measurements
2. Find the eigenvalues and eigenvectors of the covariance matrix
3. Rank order the eigenvalues to identify the first, second and subsequent principal axes
4. Use the matrix of eigenvectors to generate the new brightness values for the pixels in each of the principal axes: y= $D^T x$

Example

$$\mathbf{C}_x = \begin{bmatrix} 1.90 & 1.10 \\ 1.10 & 1.10 \end{bmatrix}$$

- To find its eigenvalues: $|\mathbf{C}_x - \lambda\mathbf{I}| = 0$

$$\begin{vmatrix} 1.90 - \lambda & 1.10 \\ 1.10 & 1.10 - \lambda \end{vmatrix} = 0$$

$$\lambda^2 - 3.00\lambda + 0.88 = 0$$

- On solving, this gives $\lambda_1 = 2.67$ and $\lambda_2 = 0.33$.

$$\mathbf{C}_y = \begin{bmatrix} 2.67 & 0 \\ 0 & 0.33 \end{bmatrix}$$

Note: sum of the eigenvalues is equal to the trace of the covariance matrix, which is simply the sum of its diagonal elements:
2.67+0.33=1.9+1.1=3

Thus the variance of the data (loosely, its scatter) in the y1 coordinate direction, which is referred to as the first principal component is 2.67 and its variance along the y2 direction, the second principal component, is 0.33. Using these values we can say that the first principal component accounts in this case for 2.67/(2.67 + 0.33) : 89% of the total variance of the data.

Example

- To find its eigenvectors: $[\mathbf{C}_x - \lambda_1 \mathbf{I}]\mathbf{g}_1 = 0$

$$\mathbf{g}_1 = \begin{bmatrix} g_{11} \\ g_{21} \end{bmatrix}$$

- Substituting for Cx and λ_1 gives the pair of equations:

$$-0.77g_{11} + 1.10g_{21} = 0$$

$$1.10g_{11} - 1.57g_{21} = 0$$

$$g_{11} = 1.43g_{21}$$

- Since D is an orthonormal matrix: $g_{11}^2 + g_{21}^2 = 1$
- So $\mathbf{g}_1 = \begin{bmatrix} 0.82 \\ 0.57 \end{bmatrix}$ and similarly $\mathbf{g}_2 = \begin{bmatrix} -0.57 \\ 0.82 \end{bmatrix}$

$$\mathbf{G} = \mathbf{D}^T = \begin{bmatrix} 0.82 & -0.57 \\ 0.57 & 0.82 \end{bmatrix}^T = \begin{bmatrix} 0.82 & 0.57 \\ -0.57 & 0.82 \end{bmatrix}$$

Example

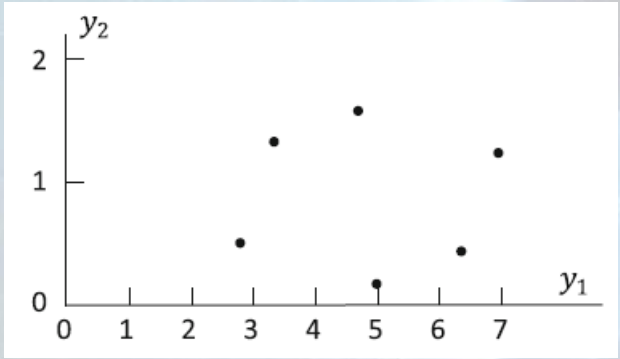
- To find the brightness values of the pixels in the principal axes:

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0.82 & 0.57 \\ -0.57 & 0.82 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 4 \end{bmatrix}, \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} 2.78 \\ 0.50 \end{bmatrix}, \begin{bmatrix} 4.99 \\ 0.18 \end{bmatrix}, \begin{bmatrix} 6.38 \\ 0.43 \end{bmatrix}, \begin{bmatrix} 6.95 \\ 1.25 \end{bmatrix}, \begin{bmatrix} 4.74 \\ 1.57 \end{bmatrix}, \begin{bmatrix} 3.35 \\ 1.32 \end{bmatrix}$$

- The pixels plotted in y space:



- <u>Notes:</u>
  - The data exhibits no obvious correlation in the new coordinates.
  - Most of the spread is in the direction of the first principal component.
  - If the pair of principal component images is generated using the y1 and y2 component brightness values for the pixels, then the first principal component will show a high degree of contrast while the second will have lower contrast by comparison.
  - The second component brightness values will be spread over a smaller part of the available range so that it appears to lack the detail of the first component.

The Effect of an Origin Shift

- It will be evident that some principal component pixel brightness values could be **negative** owing to the fact that the transformation is **a simple axis rotation**.
- Clearly, a combination of positive and negative brightness values cannot be displayed. Nor can negative brightness pixels be ignored since their appearance relative to other pixels serve to define detail.
- In practice, the problem with negative values is handled by **shifting the origin** of the principal components space so that all components have positive, and thus displayable, brightness values.
- The transformation can be done by inserting an origin shift term in the definition of the covariance matrix in the principal component space:

Let $\mathbf{y}' = \mathbf{y} - \mathbf{y}_0$ in which $\mathbf{y}_0$ is the position of a new origin. In the new $\mathbf{y}'$ coordinates

$$\mathbf{C}_{y'} = \mathcal{E}\{(\mathbf{y}' - \mathbf{m}_{y'})(\mathbf{y}' - \mathbf{m}_{y'})^{\mathrm{T}}\}$$

- The origin shift has no influence on the covariance (and thus correlation) of the data in the principal components axes: $\mathbf{C}_{y'} = \mathbf{C}_y$ *why??*

$$\mathbf{m}_{y'} = \mathbf{m}_y - \mathbf{y}_0 \quad \text{so that} \quad \mathbf{y}' - \mathbf{m}_{y'} = \mathbf{y} - \mathbf{y}_0 - \mathbf{m}_y + \mathbf{y}_0 = \mathbf{y} - \mathbf{m}_y$$

Determining the Number of Principal Components

- How many principal components do we need??

→ depends on **how much total variance** we want to keep.

	Component p						
	1	2	3	4	5	6	7
Eigenvalues, λ_p	1,010.92	131.20	37.60	6.73	3.95	2.17	1.24
Difference	879.72	93.59	30.88	2.77	1.77	.93	--
Total Variance = 1193.81							

Percent of total variance in the data explained by each component:

Computed as $\%_p = \dfrac{\text{eigenvalue } \lambda_p \times 100}{\sum_{p=1}^{7} \text{eigenvalue } \lambda_p}$.

For example,

$$\sum_{p=1}^{7} \lambda_p = 1{,}010.92 + 131.20 + 37.60 + 6.73 + 3.95 + 2.17 + 1.24 = 1{,}193.81 .$$

Percentage of variance explained by first component $= \dfrac{1{,}010.92 \times 100}{1{,}193.81} = 84.68$.

Percentage	84.68	10.99	3.15	0.56	0.33	0.18	0.10
Cumulative	84.68	95.67	98.82	99.38	99.71	99.89	100.00

- The first PC (eigenvalue 1) accounts for 84.68% of the variance.
- Component 2 accounts for 10.99% of the remaining variance.
- Cumulatively, these first two principal components account for 95.67% of the variance.

What the New Principal Components Represent?

- By computing the correlation of each *band k* with each *component p*, it is possible to determine how each band **"loads" or is associated with** each principal component.
  - Note *k* in general is the feature vector's original dimension
- Computation is done using:

$$R_{kp} = \frac{a_{kp} \times \sqrt{\lambda_p}}{\sqrt{\mathrm{Var}_k}}$$

a_{kp} = eigenvector value for band *k* and component *p* (known as **factor scores**)
p = *p*th eigenvalue
Var_k = variance of band *k* in the covariance matrix

Eigenvectors (a_{kp}) (***factor scores***):

a_{24}

	Component p						
	1	2	3	4	5	6	7
band$_k$ 1	0.205	0.637	0.327	–0.054	0.249	–0.611	–0.079
2	0.127	0.342	0.169	–0.077	0.012	0.396	0.821
3	0.204	0.428	0.159	–0.076	–0.075	0.649	–0.562
4	0.443	–0.471	0.739	0.107	–0.153	–0.019	–0.004
5	0.742	–0.177	–0.437	–0.300	0.370	0.007	0.011
7	0.376	0.197	–0.309	–0.312	–0.769	–0.181	0.051
6	0.106	0.033	–0.080	0.887	0.424	0.122	0.005

Variance–Covariance Matrix

1	100.93						
2	56.60	34.14					
3	79.43	46.71	68.83				
4	61.49	40.68	69.59	248.40			
5	134.27	85.22	141.04	330.71	568.84		
7	90.13	55.14	86.91	148.50	280.97	154.92	
6	23.72	14.33	22.92	43.62	78.91	42.65	17.78

What the New Principal Components Represent?

Degree of correlation, R_{kp}, between each band *k* and each principal component *p*:
(known as: ***factor loadings***)

	Component *p*						
	1	2	3	4	5	6	7
Band$_k$ 1	0.649	0.726	0.199	–0.014	0.049	–0.089	–0.008
2	0.694	0.670	0.178	–0.034	0.004	0.099	0.157
3	0.785	0.592	0.118	–0.023	–0.018	0.115	–0.075
4	0.894	–0.342	0.287	0.017	–0.019	–0.002	–0.000
5	0.989	–0.084	–0.112	–0.032	0.030	0.000	0.000
7	0.961	0.181	–0.152	0.065	–0.122	–0.021	0.004
6	0.799	0.089	–0.116	0.545	0.200	0.042	0.001

Computed: $R_{kp} = \dfrac{a_{kp} \times \sqrt{\lambda_p}}{\sqrt{\mathrm{Var}_k}}$.

For example:

$$R_{1,1} = \frac{0.205 \times \sqrt{1{,}010.92}}{\sqrt{100.93}} = \frac{0.205 \times 31.795}{10.046} = 0.649$$

$$R_{5,1} = \frac{0.742 \times \sqrt{1{,}010.92}}{\sqrt{568.84}} = \frac{0.742 \times 31.795}{23.85} = 0.989$$

$$R_{2,2} = \frac{0.342 \times \sqrt{131.20}}{\sqrt{34.14}} = \frac{0.342 \times 11.45}{5.842} = 0.670\,.$$

- Band 1 Blue (0.45 - 0.52 µm) 30 m
- Band 2 Green (0.52 - 0.60 µm) 30 m
- Band 3 Red (0.63 - 0.69 µm) 30 m
- Band 4 Near-Infrared (0.77 - 0.90 µm) 30 m
- Band 5 Short-wave Infrared (1.55 - 1.75 µm) 30 m
- Band 6 Thermal (10.40 - 12.50 µm) 60 m Low Gain / High Gain
- Band 7 Mid-Infrared (2.08 - 2.35 µm) 30 m

- For example, the highest correlations (i.e., factor loadings) for principal component 1 were for bands 4, 5, and 7 (0.894, 0.989, and 0.961 respectively)→ This suggests that this component is a near- and middle-infrared reflectance band.
- Principal component 2 has high loadings only in the visible bands 1, 2, and 3 (0.726, 0.670, and 0.592)→ This is a visible spectrum component.

Use of Principal Components for Image Compression

- In such a situation only the uppermost significant components are retained to represent the image.
- The **information content lost** is indicated by the sum of the eigenvalues of the components ignored compared with the total sum.
- The original image can then be reconstructed using the inverse transformation.
  - The inverse is an easy operation since “inverse=transpose”
- Generally, the image is compressed **in each band separately**.
- To **compress** the image: compressed_image=D^T original_image

  [number of PC’s*feature dimension] [feature dimension*other image dimension]

  =[number of PC’s*other image dimension]
  - where feature dimension is chosen as the number of rows or columns of the image.
- To **decompress** the image: original_image =D compressed_image

  [feature dimension*number of PC’s] [number of PC’s*other image dimension]

  =[feature dimension *other image dimension]=dimensions of original image ✓

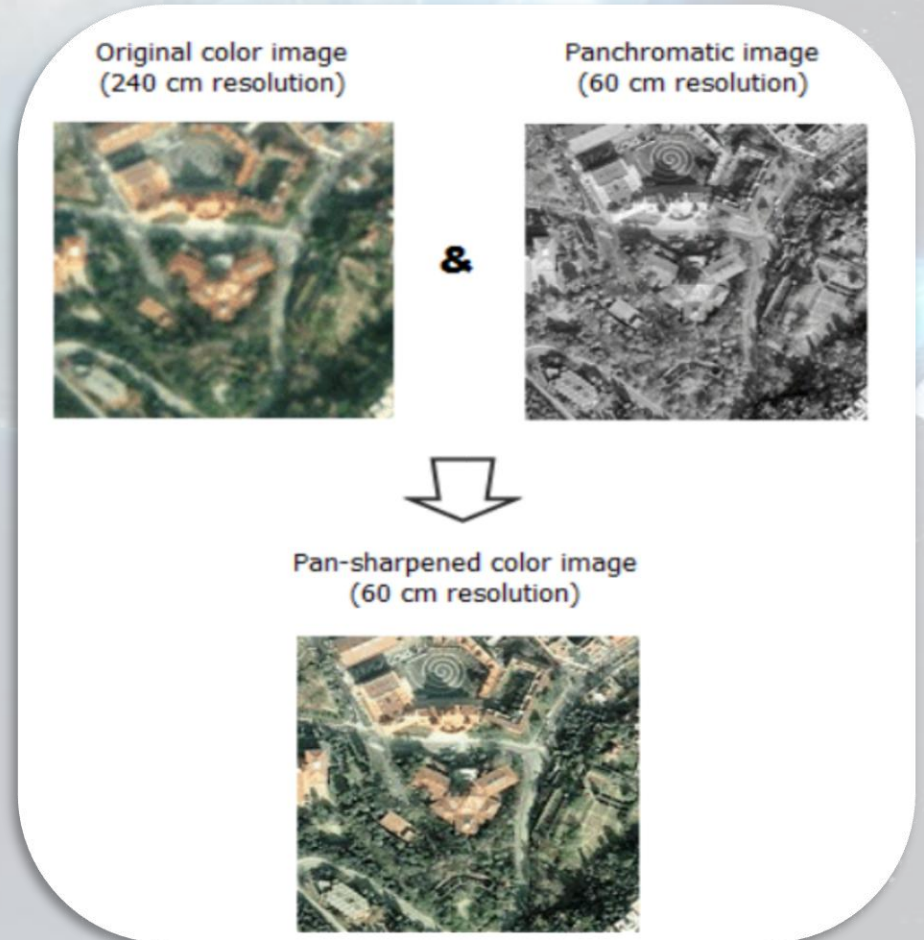

Code Time

Pan Sharpening

- A **panchromatic image** consists of only one band. It is usually displayed as a **grey scale image**.
- Pan Sharpening uses a **higher-resolution panchromatic** image (or raster band) to ***fuse*** with a **lower-resolution multiband** raster dataset.

→The result is a **multiband raster dataset with the resolution of the panchromatic raster.**

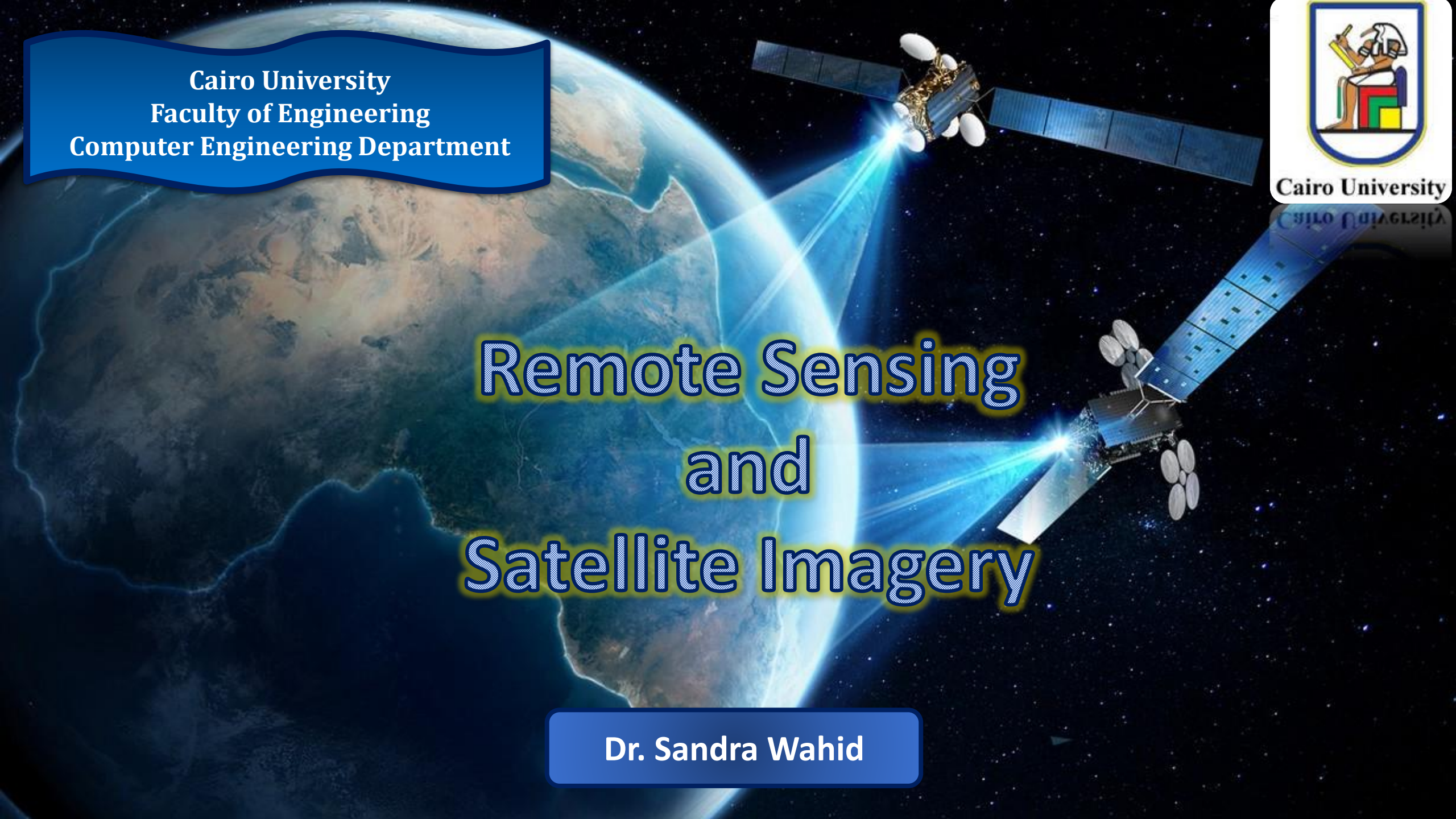
High-res grayscale band
+
Low-res color bands
=
High-res color image



- **Merging (fusing)** remotely sensed data obtained using different remote sensors must be performed carefully.
- All datasets to be merged must be accurately registered to one another and resampled to the same pixel size.


Thank You

Cairo University
Faculty of Engineering
Computer Engineering Department

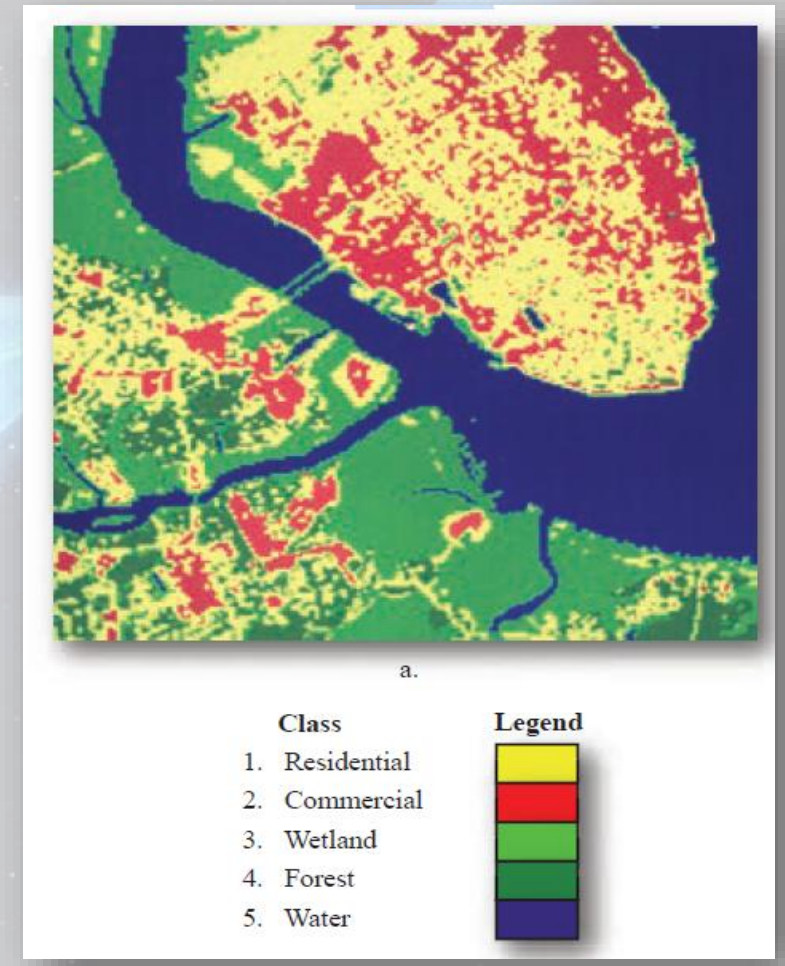


Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Classification

- Segmenting the spectral domain into regions that can be associated with the **land-cover types/classes** of interest to a particular application (e.g., urban, agriculture, or wetland). In practice those regions may sometimes overlap.
- Known as extraction of thematic information→ used to build a **thematic map** from the thematic map classes (e.g., water or forest).
- Classification can be either:
  - **Supervised:** available labelled training data
  - **Unsupervised:** no labelled training data
- Classification can be either:
  - **Hard (crisp):** classify to one specific class
  - **Soft (fuzzy):** generate membership values for all classes (e.g., 10% soil, 10% sand, 80% forest)
- Classification can be either:
  - **Per-pixel:** processing the entire scene pixel by pixel
  - **Object-based:** first the image is segmented into discrete objects and then each object is classified



Supervised Classification

- Essential practical steps in applying a classifier:

1. Deciding on the classes of interest.
2. Collecting training data: choosing known, representative pixels for each of the classes. Those pixels are said to form the training data. Sometimes the training pixels for a given class will lie in a common region enclosed by a border. That region is called a ***training field/site/area***.
3. Using the training data to estimate the parameters of the classifier algorithm; the set of parameters is sometimes called the ***signature*** of that class.
4. Using the trained classifier to label every pixel/object in the image.
5. Producing thematic (class) maps and tables which summarize class memberships of all pixels/objects in the image.
6. Assessing the accuracy of the final product using a labelled testing data set.

Feature Selection

- Determine the bands that are most effective in discriminating each class from all others.
  - Can use a **portion or all** the bands that exist in the dataset.
  - Can also use **composite** bands (such as NDVI).
  - Can also use **dimensionality reduction** (such as PCA) to obtain new synthetic bands.

- There are methods for feature selection where **graphical** methods can be followed by **statistical** methods.

- In any case, we will deal with a **pixel vector** that represents our feature vector for each pixel.

Supervised Classification

- Various supervised classification algorithms can be used to assign an unknown pixel to one of *m* possible classes.
- Classification algorithms can be:
  - **Parametric:** assume normally distributed data and knowledge about the forms of the underlying class density functions.

  ➢The most widely adopted parametric classification algorithms is:
    - The maximum likelihood.
  - **Nonparametric:** make no such assumption.

  ➢ Several widely adopted nonparametric classification algorithms include:
    - One-dimensional density slicing
    - Parallepiped
    - Minimum distance
    - Nearest-neighbors
    - Neural networks
    - Expert systems

Maximum Likelihood Classification

Bayes' Classification:

- Let the classes be represented by $\omega_i, i = 1 \ldots M$ where M is the number of classes
- The vector ***x*** is a column vector of the brightness values for the pixel in each measurement band→ the pixel is a point in spectral space.
- The probability $p(w_i|\boldsymbol{x})$ tells us the likelihood that w_i is the correct class for the pixel at position ***x*** in the spectral space.
- If we knew $p(w_i|\boldsymbol{x})$ for all classes w_i then we could label the pixel (classify it) according to the decision rule: $\mathbf{x} \in \omega_i \text{ if } p(\omega_i|\mathbf{x}) > p(\omega_j|\mathbf{x}) \text{ for all } j \neq i$
  - Known as: maximum posterior or MAP decision rule.

The Maximum Likelihood Decision Rule:

- Since $p(w_i|\boldsymbol{x})$ are unknown, use Bayes' theorem: $p(\omega_i|\mathbf{x}) = p(\mathbf{x}|\omega_i)\,p(\omega_i)/p(\mathbf{x})$
- The decision rule becomes: $\mathbf{x} \in \omega_i \text{ if } p(\mathbf{x}|\omega_i)p(\omega_i) > p(\mathbf{x}|\omega_j)p(\omega_j) \text{ for all } j \neq i$
  - $p(\boldsymbol{x})$ has been removed as a common factor, since it is not class dependent.

Maximum Likelihood Classification

Multivariate Normal Class Models:

- To develop the maximum likelihood classifier further we now choose a particular **probability model** for the class conditional density function $p(\mathbf{x}|w_i)$

→The most common choice is a **multivariate normal distribution**, also called a **Gaussian distribution**.

- For an N dimensional space the specific form of the multivariate Gaussian distribution function: $$p(\mathbf{x}|\omega_i) = (2\pi)^{-N/2}|\mathbf{C}_i|^{-1/2}\exp\left\{-\tfrac{1}{2}(\mathbf{x}-\mathbf{m}_i)^{\mathrm{T}}\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i)\right\}$$
  - where $\boldsymbol{m}_i$ and $\boldsymbol{C}_i$ are the mean vector and covariance matrix of the data in class w_i
  - $|\boldsymbol{C}_i|$ is the determinant
  - shorthand form: $\mathcal{N}(\mathbf{x}|\mathbf{m}_i, \mathbf{C}_i)$
- In log space, define the ***discriminant function***: $$g_i(\mathbf{x}) = \ln\{p(\mathbf{x}|\omega_i)p(\omega_i)\} = \ln p(\mathbf{x}|\omega_i) + \ln p(\omega_i)$$
- The decision rule becomes: $$\mathbf{x} \in \omega_i \text{ if } g_i(\mathbf{x}) > g_j(\mathbf{x}) \text{ for all } j \neq i$$
- By substitution: $$g_i(\mathbf{x}) = -\tfrac{1}{2}N\ln 2\pi - \tfrac{1}{2}\ln|\mathbf{C}_i| - \tfrac{1}{2}(\mathbf{x}-\mathbf{m}_i)^{\mathrm{T}}\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i) + \ln p(\omega_i)$$

Maximum Likelihood Classification

Multivariate Normal Class Models:

$$g_i(\mathbf{x}) = -\tfrac{1}{2}N\ln 2\pi - \tfrac{1}{2}\ln|\mathbf{C}_i| - \tfrac{1}{2}(\mathbf{x}-\mathbf{m}_i)^{\mathrm{T}}\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i) + \ln p(\omega_i)$$

- Removing the first term since it is not class dependent:

$$g_i(\mathbf{x}) = \ln p(\omega_i) - \tfrac{1}{2}\ln|\mathbf{C}_i| - \tfrac{1}{2}(\mathbf{x}-\mathbf{m}_i)^{\mathrm{T}}\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i)$$

- If there is no useful information about the values of the **prior probabilities** ***p(w$_i$)***, they are assumed all to be equal:

$$g_i(\mathbf{x}) = -\ln|\mathbf{C}_i| - (\mathbf{x}-\mathbf{m}_i)^{\mathrm{T}}\mathbf{C}_i^{-1}(\mathbf{x}-\mathbf{m}_i)$$

This is the discriminant function for **the Gaussian maximum likelihood classifier**, so-called because it essentially determines class membership of a pixel based on **the highest of the class conditional probabilities, or likelihoods**.

Maximum Likelihood Classification

Decision Surfaces:

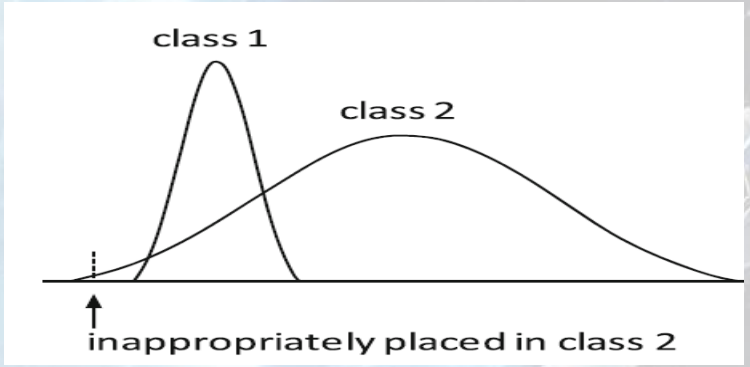
- So far, the probability distributions have been used to discriminate among the candidate class labels for a pixel.
- We can also find the shapes of the **separating boundaries** in vector space which allows us to assess the relative classification abilities, or strengths, of different classifiers.
- If classes are defined by regions in spectral space where their discriminant functions are the largest →those regions are separated by surfaces where the discriminant functions for adjoining classes are **equal**. The ith and jth classes are separated by the surface: $g_i(\mathbf{x}) - g_j(\mathbf{x}) = 0$

Maximum Likelihood Classification

Thresholds:

- Pixels at every point in spectral space will be classified into one of the available classes w_i; irrespective of how small the actual probabilities of class membership are.
- This may result in poor classification,

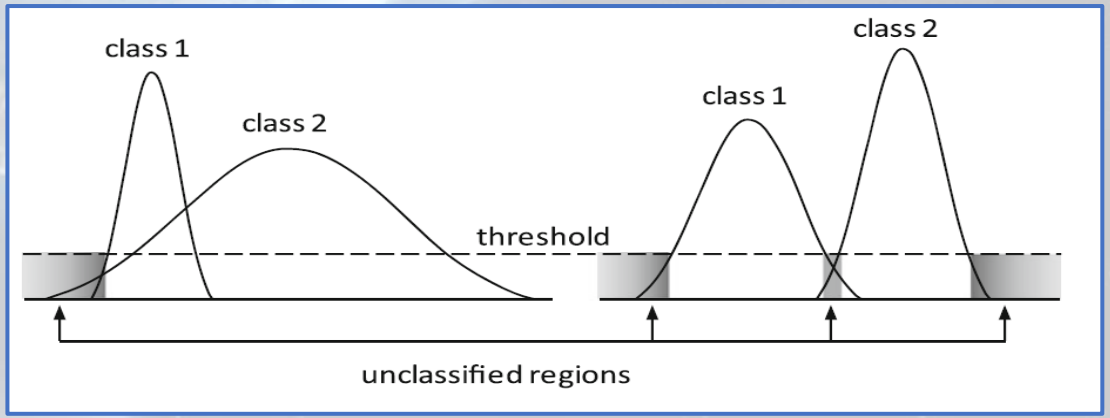
  example for one dimensional data:



- Solution: use thresholds.
  - Pixels which have probabilities for all classes below the threshold are **not classified**.

$$\mathbf{x} \in \omega_i \quad \text{if} \quad g_i(\mathbf{x}) > g_j(\mathbf{x}) \text{ for all } j \neq i$$
$$\text{and } g_i(\mathbf{x}) > T_i$$

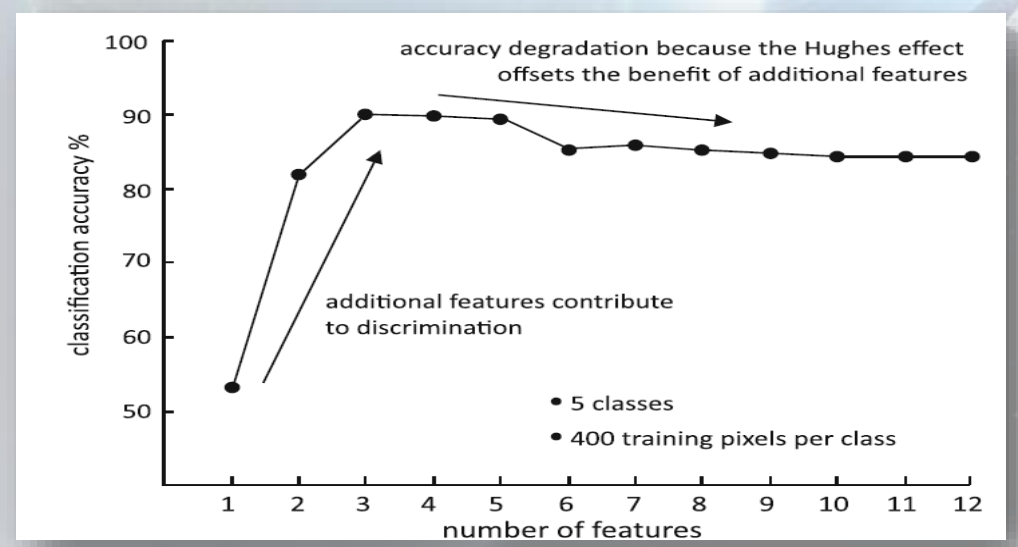
where T_i is the threshold to be used on class w_i



Maximum Likelihood Classification

The Hughes Phenomenon and the Curse of Dimensionality:

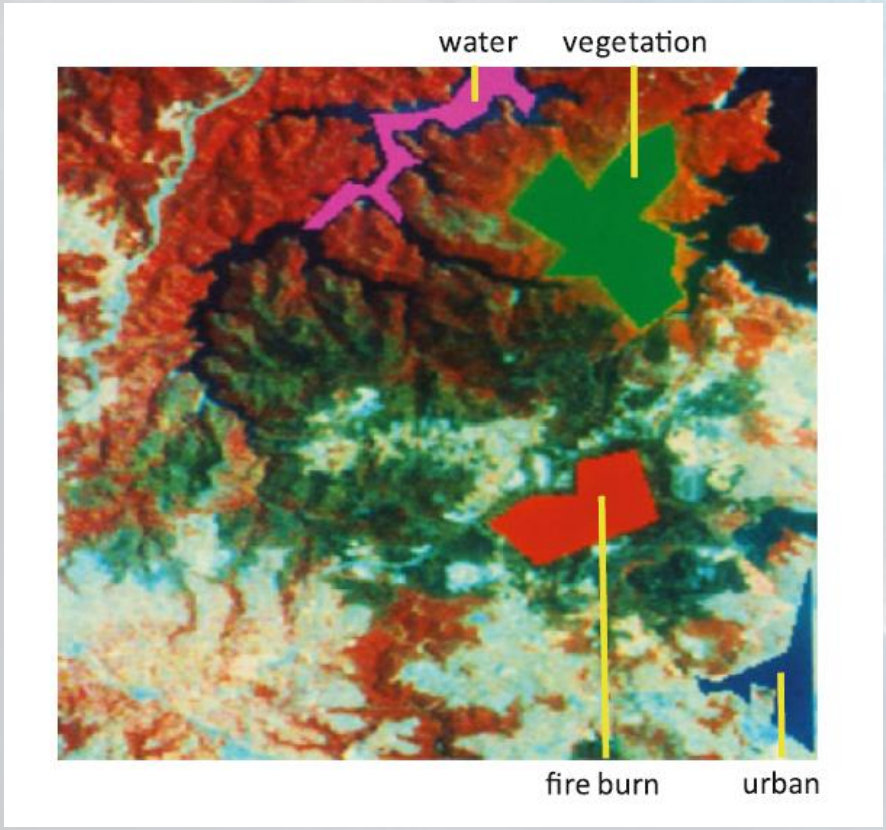
- Suppose we have enough training data for a given number of dimensions (features) to provide reliable training (good **estimates of the class parameters**) and good **generalization** (the extent to which a classifier predicts on unseen data).
- Adding more features or dimensions suggests that the results should be improved because more features should bring more information into the training phase.
- However, if we don't increase the number of training samples sufficiently, we may not be able to get good estimates of the larger set of parameters created by the increased dimensionality, and both training and generalization will suffer →That is the essence of the Hughes phenomenon.



Maximum Likelihood Classification

Example:

- Image segment of 256*276 pixels to be classified.
- Four training areas/fields are identified in solid color:
  - water (violet), vegetation (green), fire burn (red) and urban (dark blue).
  - Those pixels (their BV's) were used to generate the class signatures:



Class	Mean vector	Covariance matrix			
Water	44.27	14.36	9.55	4.49	1.19
	28.82	9.55	10.51	3.71	1.11
	22.77	4.49	3.71	6.95	4.05
	13.89	1.19	1.11	4.05	7.65
Fire burn	42.85	9.38	10.51	12.30	11.00
	35.02	10.51	20.29	22.10	20.62
	35.96	12.30	22.10	32.68	27.78
	29.04	11.00	20.62	27.78	30.23
Vegetation	40.46	5.56	3.91	2.04	1.43
	30.92	3.91	7.46	1.96	0.56
	57.50	2.04	1.96	19.75	19.71
	57.68	1.43	0.56	19.71	29.27
Urban	63.14	43.58	46.42	7.99	−14.86
	60.44	46.42	60.57	17.38	−9.09
	81.84	7.99	17.38	67.41	67.57
	72.25	−14.86	−9.09	67.57	94.27

Parallelepiped Classification

- The classification decision rule is based on simple **Boolean "and/or"** logic.
- The **mean (μ_{ck})** and **standard deviation (σ_{ck})** is computed for each class *c* for each band *k*.
- A parallelepiped algorithm decides Bv_{ijk} is in class *c* if, and only if:

$$\mu_{ck} - \sigma_{ck} \leq BV_{ijk} \leq \mu_{ck} + \sigma_{ck}$$

where

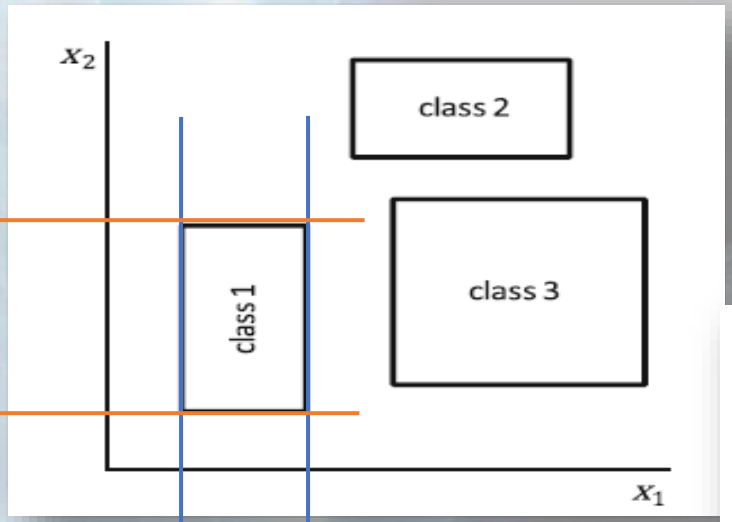
$c = 1, 2, 3, \ldots, m$, number of classes, and
$k = 1, 2, 3, \ldots, n$, number of bands.

- The low and high decision boundaries are defined as: $L_{ck} = \mu_{ck} - \sigma_{ck}$ $H_{ck} = \mu_{ck} + \sigma_{ck}$
- The parallelepiped algorithm becomes:

$$L_{ck} \leq BV_{ijk} \leq H_{ck}$$

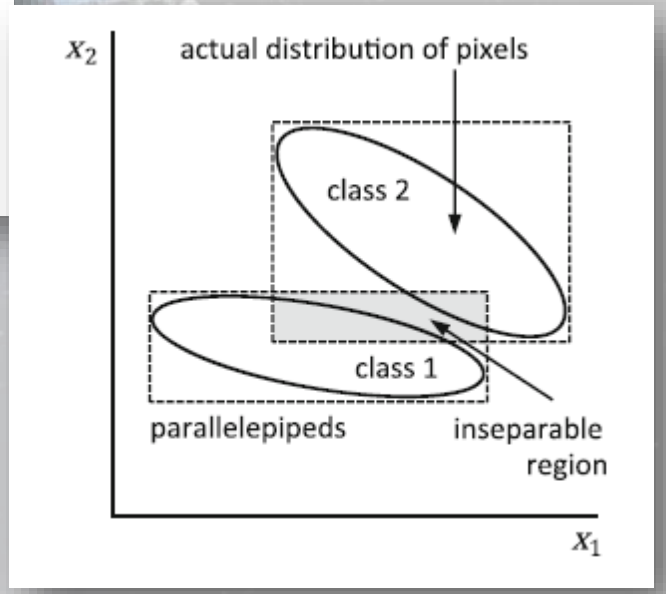
Parallelepiped Classification

- These decision boundaries form an *n*-dimensional box/parallelepiped in feature space.
- Unknown pixels are labelled as coming from the class of the parallelepiped within which they lie→ If the pixel value lies above the low threshold and below the high threshold for all n bands evaluated, it is assigned to that class.
- A 2-dimensional case:



- It is a **very simple** and **fast** classifier.
- <u>Limitations:</u>

1. There can be considerable gaps between the parallelepipeds in spectral space
   →pixels in those regions cannot be classified.
2. For correlated data some parallelepipeds can overlap:
   →pixel are usually assigned to the first class for which it meets all criteria.



Parallelepiped Classification

- Example:

- Pixel ***a*** has a value of 40 in both bands 4 and 5.
- Pixel ***b*** has a value of 10 in band 4 and 40 in band 5.

Class	Lower Threshold, L_{ck}	Upper Threshold, H_{ck}	Does pixel *a* (40, 40) satisfy criteria for this class in this band? $L_{ck} \leq a \leq H_{ck}$	Does pixel *b* (10, 40) satisfy criteria for this class in this band? $L_{ck} \leq b \leq H_{ck}$
1.Residential Band 4 Band 5	36.7 – 4.53 = 31.27 55.7 – 10.72 = 44.98	36.7 + 4.53 = 41.23 55.7 + 10.72 = 66.42	Yes No	No No
2.Commercial Band 4 Band 5	54.8 – 3.88 = 50.92 77.4 – 11.16 = 66.24	54.8 + 3.88 = 58.68 77.4 + 11.16 = 88.56	No No	No No
3.Wetland Band 4 Band 5	20.2 – 1.88 = 18.32 28.2 – 4.31 = 23.89	20.2 + 1.88 = 22.08 28.2 + 4.31 = 32.51	No No	No No
4.Forest Band 4 Band 5	39.1 – 5.11 = 33.99 35.5 – 6.41 = 29.09	39.1 + 5.11 = 44.21 35.5 + 6.41 = 41.91	Yes Yes, assign pixel to class 4, forest. STOP.	No Yes
5.Water Band 4 Band 5	9.3 – 0.56 = 8.74 5.2 – 0.71 = 4.49	9.3 + 0.56 = 9.86 5.2 + 0.71 = 5.91	— —	No No, assign pixel to un-classified category. STOP.

Assessing Classification Accuracy

- The Error Matrix (contingency matrix or a confusion matrix):

thematic map classes \ reference data classes	A	B	C	sum
A	35	2	2	39
B	10	37	3	50
C	5	1	41	47
sum	50	40	46	136

overall accuracy =(35+37+41)/136≡83.1%

	producer's accuracies		user's accuracies
A	35/50≡70.0%	A	35/39≡89.7%
B	37/40≡92.5%	B	37/50≡74.0%
C	41/46≡89.1%	C	41/47≡87.2%

- **Errors of omission** correspond to those pixels belonging to the reference class that the classifier has failed to recognize.
- **Errors of commission** correspond to those pixels belonging to other reference classes that the classifier has placed in the class of interest.

- **Omission errors:**
  - A: 15/50
  - B: 3/40
  - C: 5/40

- **Commission errors:**
  - A: 4/39
  - B: 13/50
  - C: 6/47

PA is the producer's accuracy (recall or true positive rate or sensitivity): $\frac{TP}{TP+FN}$,

UA the user's accuracy (consumer's accuracy or precision or positive predictive value): $\frac{TP}{TP+FP}$,

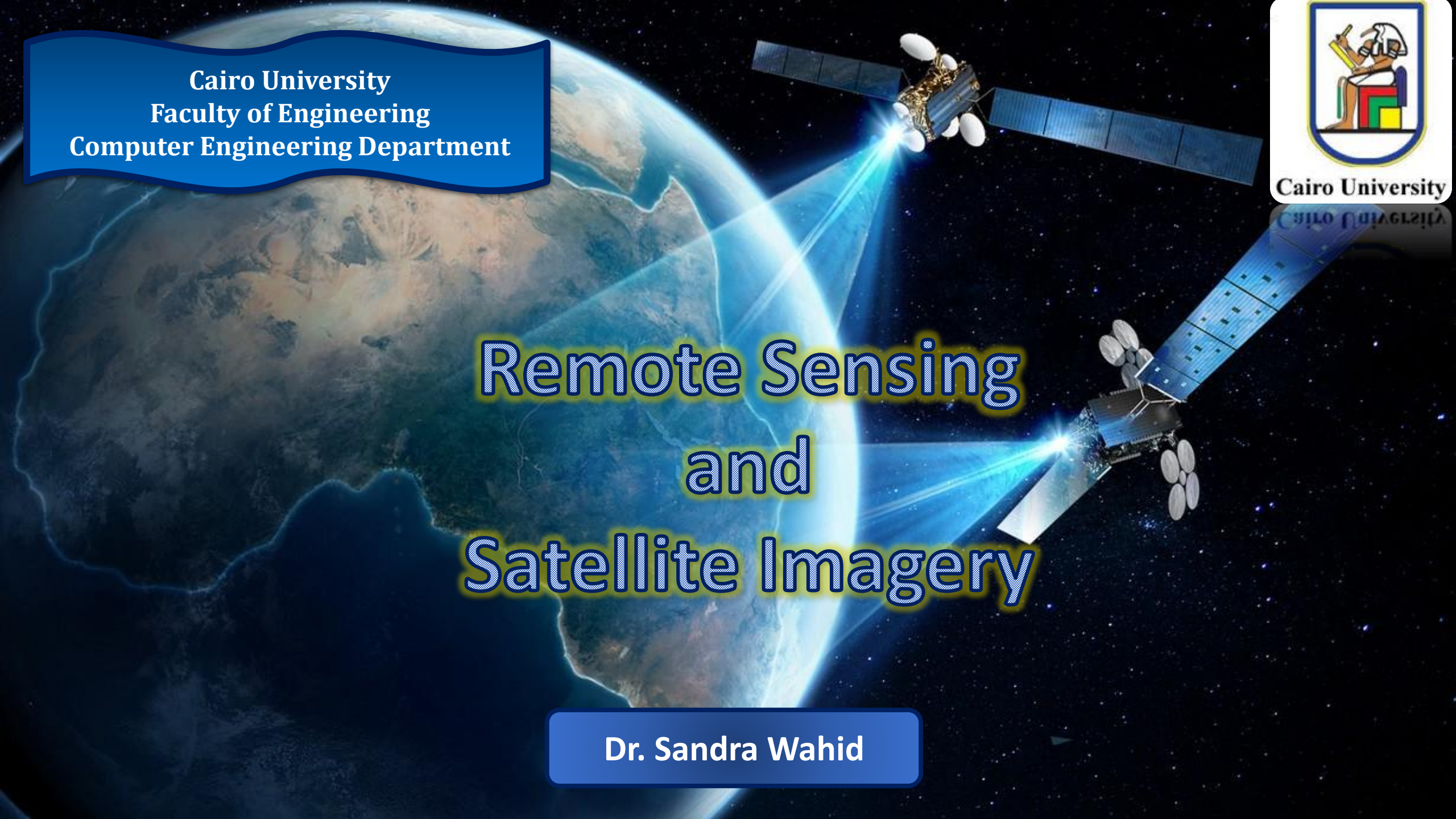
OA the (overall) accuracy: $\frac{TP+TN}{TP+FP+FN+TN}$,

OE omission error (false negative rate): $1-PA=\frac{FN}{TP+FN}$,

CE commission error (false discovery rate): $1-UA=\frac{FP}{TP+FP}$,


Thank You

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Remote Sensing and Satellite Imagery

Dr. Sandra Wahid

Unsupervised Classification

- When **labelled training data is not available** and yet it would still be of interest to convert remote sensing image data into a thematic map of labels.

- Such an approach is called unsupervised classification since the analyst, in principle, takes no part in an algorithm's learning process.

- Several methods are available for unsupervised learning, the most common in remote sensing is based on the use of **clustering algorithms**, which seek to identify pixels in an image that are **spectrally similar**.

ISODATA Method

- A widely used clustering algorithm is the Iterative Self-Organizing Data Analysis Technique (ISODATA).

- The ISODATA algorithm is a **modification of the k-means** clustering algorithm, which includes:
  - a) **merging** clusters if their separation distance in the feature space is below a user-specified threshold
  - b) rules for **splitting** a single cluster into two clusters

- Don’t need to know the number of clusters (as in k-means where k is the number of clusters).

- User **defines threshold** values for parameters.

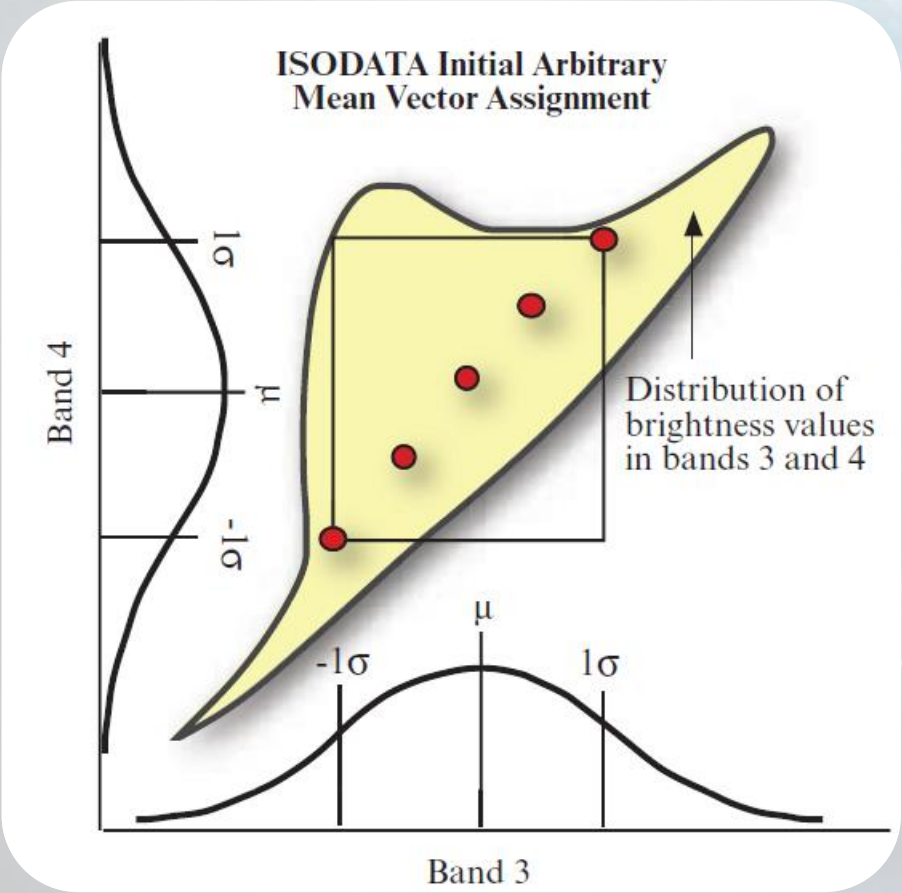
- ISODATA is **iterative** because it makes a large number of passes through the dataset until specified results are obtained

User-defined Thresholds in ISODATA

- ***C_{max}***: the maximum number of clusters→ It is common that fewer are found in the final classification map after splitting and merging take place.
- ***T***: the percentage of pixels whose clusters remain unchanged between iterations→ When this number is reached, the ISODATA algorithm terminates (convergence).
- ***M***: the maximum number of iterations.
- ***Minimum members in a cluster (%)***: If a cluster contains less than this %, it is deleted and the members are assigned to an alternative cluster **→Clusters Deletion**
  - The default is often set to 1%.
- ***Maximum standard deviation (σ_{max})***: When the standard deviation (σ) for a cluster exceeds σ_{max} and the number of members in the cluster is greater than twice the specified minimum members in a cluster, the cluster is split into two clusters. The mean vectors for the two new clusters are the old cluster centers ± 1σ **→Clusters Split**
  - Very typical values are 4.5 and 7.
- ***Split separation value***: If specified (not 0), it takes the place of the standard deviation in determining the locations of the new mean vectors plus and minus the split separation value.
- ***Minimum distance between cluster means (C)***: Clusters with a distance less than this value are merged **→Clusters Merge**
  - The default is often set to 3.0.

ISODATA Initial Arbitrary Cluster Allocation

- An initial arbitrary assignment of all C_{max} mean vectors takes place by choosing these means in a certain region in the feature space: this region is defined by the features' parallelepiped.
- Consider a two-dimensional example using bands 3 and 4:



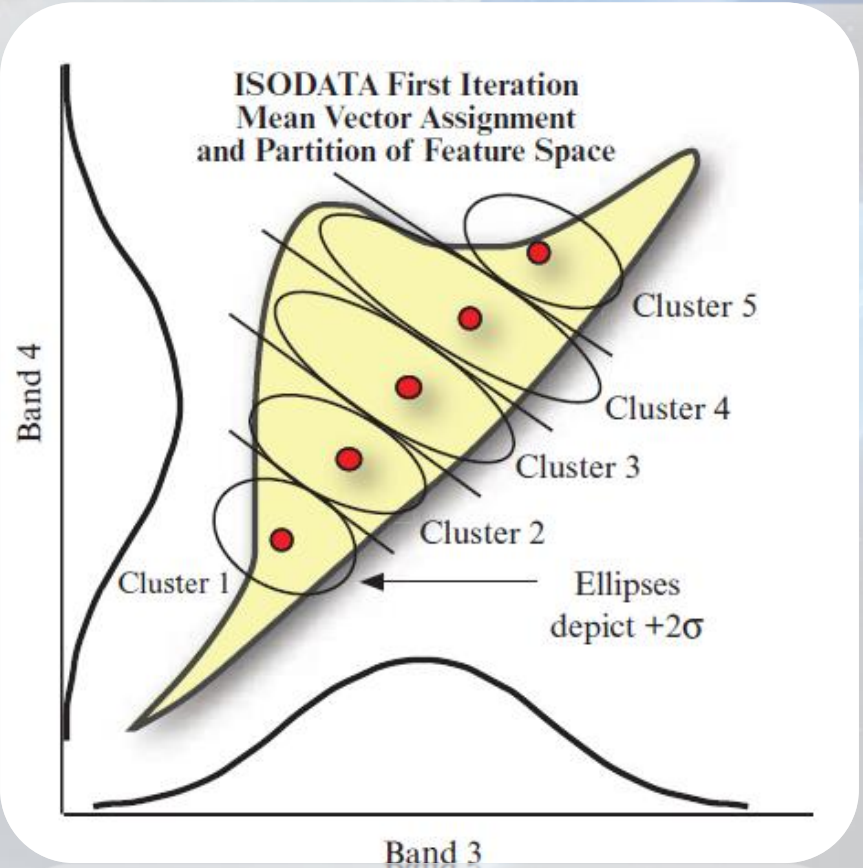
Five mean vectors are distributed along the vector beginning at location μ_3-σ_3, μ_4-σ_4 and ending at μ_3+σ_3, μ_4+σ_4 (chosen uniformly spaced)

The fact that the initial mean vectors are situated throughout the heart of the existing data is superior to random initiation.

ISODATA First Iteration

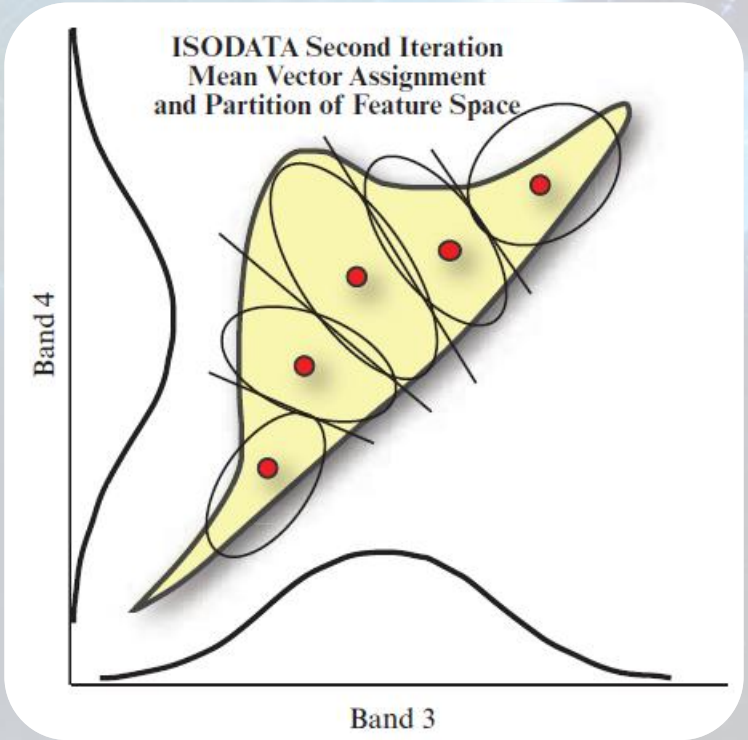
- Each candidate pixel is compared to each cluster mean and assigned to the cluster whose mean is closest using an appropriate distance metric.
  - **Euclidean distance** is commonly used:

$$Euc(A, B) = \sqrt{\sum_{i=1}^{n}(A_i - B_i)^2}$$



ISODATA Second to Mth Iteration

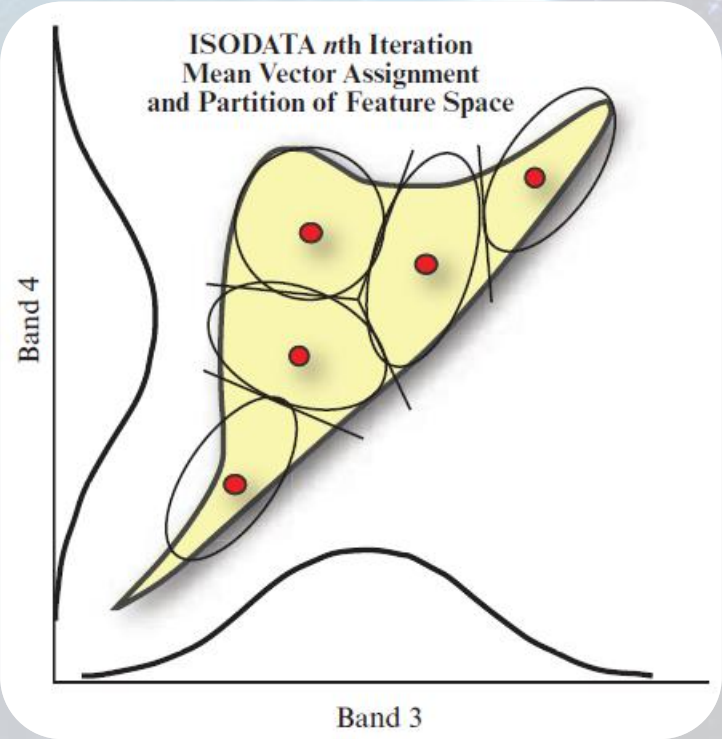
- After the first iteration, a new mean for each cluster is calculated based on the actual spectral locations of the pixels assigned to each cluster, instead of on the initial arbitrary calculation → *till now same as k-means clustering*
- This involves analysis of the following parameters to ***merge or split*** clusters:
  - Minimum members in a cluster (%) **→Clusters Deletion**
  - Maximum standard deviation (σ_{max}) and split separation value **→Clusters Split**
  - Minimum distance between cluster means (C) **→Clusters Merge**



ISODATA Second to Mth Iteration

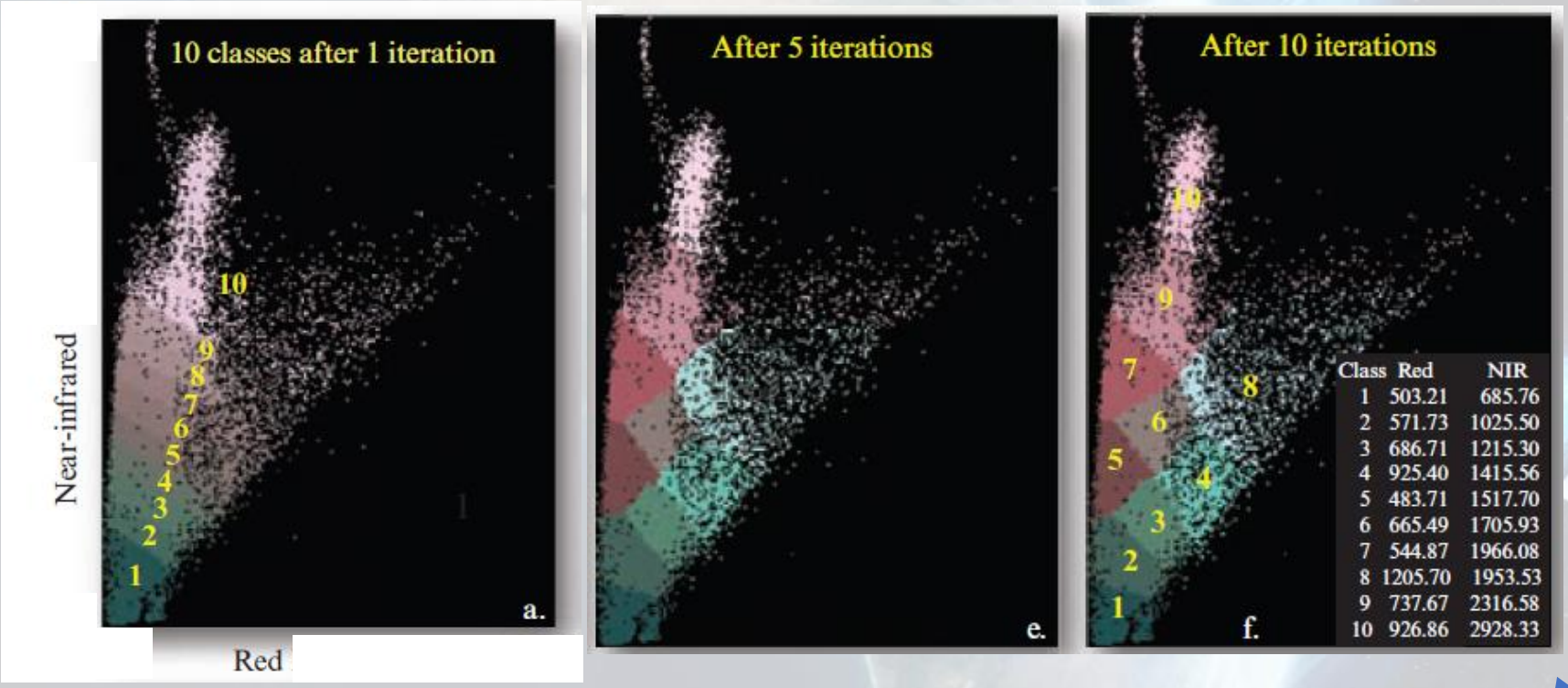
- Then the entire process is repeated with each candidate pixel once again compared to the new cluster means and assigned to the nearest cluster mean.
- This iterative process continues until there is
  - 1) little change in cluster assignment between iterations (**T threshold is reached**)

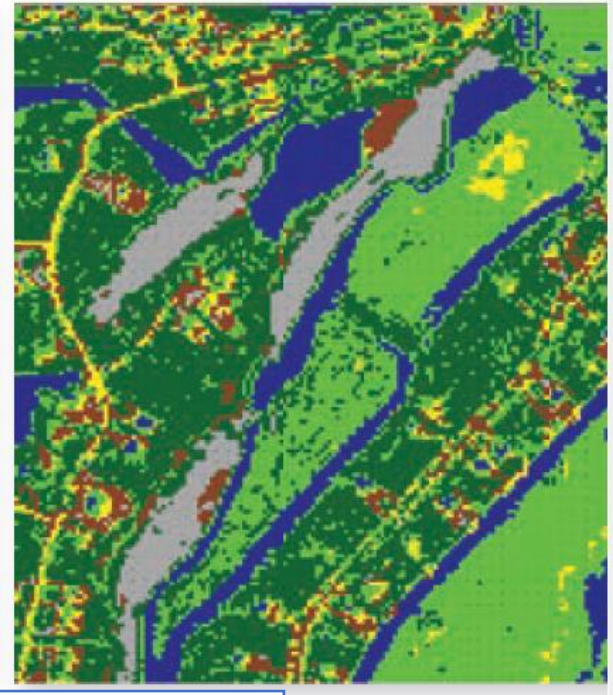
  or
  - 2) the maximum number of iterations is reached (**M**).



ISODATA Example

- Two bands (red and near-infrared).
- Only 10 clusters for demonstration purposes.





Classification Map

Class	Legend
1. Water	
2. Wetland	
3. Wetland	
4. Roof/asphalt	
5. Forest	
6. Wetland	
7. Forest	
8. Bare soil	
9. Forest	
10. Fairway	

Finally, a **classification map** is derived from the computed clusters → If some pixels within a given cluster can be identified with a particular land-cover type (by means of maps, site visits or other forms of reference data, e.g., presence of rivers or roads) then all pixels within the same cluster can be assumed to be from that class. This method of image classification depends on **a posteriori** recognition of the classes.

Unsupervised Cluster Busting

- It is common when performing unsupervised classification to generate *n* clusters (e.g., 100) that we have **no confidence in labeling** *q* of them to an appropriate land-cover class (e.g., 30) known as "**mixed clusters**"

  → When this occurs, it may be possible to perform **cluster busting**.

- First, all the pixels associated with the *q* clusters that are difficult to label are all recoded to a value of 1 and a binary mask file is created. A mask program is then run to generate a new multiband image file consisting of only the pixels that could not be adequately labeled during the initial unsupervised classification.

- Then a new unsupervised classification is performed on this new image.

- Usually, there are still some clusters that contain mixed pixels, but the proportion definitely goes down.

- The process may be repeated one more time to see if an additional unsupervised classification breaks out additional clusters.

Assessing Performance

- If the task at hand has some labelled data, then you can use it for accuracy assessment as in **supervised classification** e.g., accuracy, F-score, …
- If it is a purely unsupervised task (no labeled data), then there are some **metrics** that can be used to evaluate the performance of clustering algorithms, for example:

Silhouette Coefficient:

$$s = \frac{b - a}{max(a, b)}$$

  - a: The mean distance between a sample and all other points in the same cluster → intra-cluster distance.
  - b: The mean distance between a sample and all other points in the next nearest cluster → inter-cluster distance.
  - Results in the range of -1 to 1.
    - A score near 1 signifies that the data point is very compact within the cluster to which it belongs and far away from the other clusters→ sample is well-clustered.
    - A score near -1 signifies that the data point isn't assigned to the right cluster.
    - A score near 0 signifies overlapping clusters.
  - The Silhouette Coefficient for a set of samples is given as the mean of the Silhouette Coefficient for each sample.


Thank You